



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

I/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

Under Choice Based Credit System

GROUP-A (CIVIL, EEE & MECHANICAL)

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1101	English – I *	3	3	1	--	4	30	70	100
B17 BS 1102	Mathematics – I *	3	3	1	--	4	30	70	100
B17 BS 1105	Engineering Chemistry	3	3	1	--	4	30	70	100
B17 ME 1101	Engineering Mechanics	3	3	1	--	4	30	70	100
B17 ME 1102	Engineering Drawing	3	1	--	3	4	30	70	100
B17 CE 1101	Environmental Studies *	2	2	1	--	3	30	70	100
B17 BS 1107	Engineering Chemistry Lab	2	--	--	3	3	50	50	100
B17 BS 1108	English Communication Skills Lab – I *	2	--	--	3	3	50	50	100
B17 BS 1109	Engineering Workshop & IT Workshop	2	--	--	3	3	50	50	100
B17 BS 1111	Inner Engineering	--	--	--	2	2	--	--	--
Total		23	15	5	14	34	330	570	900

* Common to both Group - A and Group - B

Code: B17 BS 1101

ENGLISH - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop the communication skills of the students in both the formal and informal situations.
4. To expose the students to various forms of writing in formal settings in order to build confidence in standard grammar
5. To introduce the students various types of reading comprehension practices.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand the rudiments of LSRW Skills, comprehension and fluency of speech.
2. Gain confidence and competency in vocabulary and grammar.
3. Listen, speak, read and write effectively in both the academic and non- academic environment.
4. Extend his/her reading skills towards literature.
5. Strengthen his/her analytical and compositional skills.

SYLLABUS

Life through Language: An Effective Learning Experience

Life through Language has a systematic structure that builds up communicative ability progressively through the chapters. It will enable the learner to manage confusion; frame question for themselves and others; develop new ideas; support ideas with evidence; express themselves with poise and clarity; and think critically. Acquisition of skill leads to confidence.

UNIT-I

People and Places:- Word search - Ask yourself-Self-assessment-I -Self-assessment-II - Sentence and its types- Describing people, places and events-Writing sentences-Self-awareness-Self-motivation, Dialogue writing.

UNIT-II

Personality and Lifestyle:- Word quiz – Verbs-Adverbs-Negotiations-Proving yourself-Meeting Carl Jung- Describing yourself- Living in the 21st century- Using your dictionary-Communication-Adaptability.

UNIT-III

Media and Environment: - A list of 100 basic words – Nouns- Pronouns- Adjectives-News report- Magazine article- User's Manual for new iPod- A documentary on the big cat- Why we need to save our tigers: A dialogue- Global warming- Paragraph Writing-Arguing a case-Motivation- Problem solving.

UNIT-IV

Entertainment and Employment:- One word substitutes- Parts of speech- Gerunds and infinitives- An excerpt from a short story an excerpt from a biography- A consultant interviewing employees- Your first interview- Reality TV- Writing an essay-Correcting sentences- Integrity Sense of humor.

UNIT-V

Work and Business:- A list of 100 difficult words- Articles, Quantifiers- Punctuation - Open letter to the Prime Minister Business dilemmas: An email exchange- A review of *IPL: The Inside Story*, Mark Zuckerberg: World's Youngest Billionaire- A conversation about a business idea- Pair work: Setting up a new business- Recession- Formal letters-Emails- Reports- Professionalism-Ethics, Fill in the blanks.

Text Book:

1. *Life through Language: A Holistic Approach to Language Learning*. Board of Editors, Pearson Publishers, India. 2013.

Reference Books:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008.
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008.
3. Know Your English (Volume 1&2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Mathukutty Monippally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that is essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Learn about linear first order and higher order ordinary differential equations and their applications.
4. Acquire knowledge of Laplace transform, partial differentiation and their applications.
5. Learn certain first order and higher order partial differential equations.

Course Outcomes: At the end of the Course, Student will be able to:

1. Solve linear ordinary differential equations of first order and first degree. Also will be able to apply the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.
2. Solve linear ordinary differential equations of second order and higher order. Also will be able to apply the knowledge in simple applications such as LCR circuits and Simple harmonic motion.
3. Determine Laplace transform and inverse Laplace transform of various functions.
4. Use Laplace transforms to solve a linear ODE.
5. Calculate total derivative, Jacobian and maxima/minima of functions of two variables.
6. Form partial differential equations and solve some standard types of first order PDEs. Find complimentary function and particular integral of linear higher order homogeneous and non-homogeneous PDEs.

SYLLABUS

UNIT I: Differential equations of first order and first degree:

Linear, Bernoulli, Exact, Reducible to exact types.

Applications: Newton's Law of cooling, Law of natural growth and decay, Orthogonal trajectories, Simple electrical circuits, Chemical reactions.

UNIT II: Linear differential equations of higher order:

Non-homogeneous equations of higher order with constant coefficients with RHS term of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax}V(x)$, $xV(x)$, Method of Variation of parameters.

Applications: LCR circuit, Simple Harmonic motion.

UNIT III: Laplace transforms:

Laplace transforms of standard functions, transforms of $tf(t)$, $f(t)/t$, properties, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function, Inverse Laplace transforms, convolution theorem (without proof).

Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms.

UNIT IV: Partial differentiation:

Introduction, Homogeneous functions, Euler's theorem, Total derivative, Chain rule, which variable is to be treated as constant, Functional dependence, Jacobians, Taylor series for a function of two variables, Leibnitz rules for differentiation under the integral sign.

Applications: Errors and Approximations, Maxima and Minima of functions of two variables without constraints, Lagrange's method (with constraints)

UNIT V: First order and higher order partial differential equations:

Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of first order Lagrange linear equation and nonlinear equations of standard types (excluding Charpit's method).

Solutions of Linear homogeneous and non-homogeneous Partial differential equations with constant coefficients - RHS terms of the type e^{ax+by} , $\sin(ax+by)$, $\cos(ax+by)$, $x^m y^n$.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
2. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson.
3. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
4. Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING CHEMISTRY
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Knowledge of basic concepts of chemistry for engineering students will help them as professional engineers later in design and material selection as well as utilizing the available resources.

Course Objectives:

1. To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose.
2. To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc.
3. To understand the concepts of galvanic cells and corrosion with theories like electro chemical theory.
4. To understand the importance of water.
5. To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries.

Course Outcomes:

1. At the end of the course the students learn the advantages and limitations of plastic materials and their use in design.
2. Fuels which are used commonly and their economics, advantages and limitations are discussed.
3. Students gained Knowledge reasons for corrosion and some methods of corrosion control.
4. Students understands the impurities present in raw water, problems associated with them and how to avoid them.
5. Similarly students understand liquid crystals and semi conductors. Students can gain the building materials , solar materials, lubricants and energy storage devices.

SYLLABUS

UNIT-I: High Polymers and Plastics; Rubbers & Elastomers:

Polymerization Definition, Types of Polymerization, Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6, Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers, Biodegradable Polymers - PHBV, Nylon 2, Nylon 6. Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N;

UNIT-II :Fuel Technology& Lubricants:

Fuels: - Introduction – Classification of fuels, Calorific value – HCV and LCV, Determination of Calorific value by bomb calorimeter; Proximate and ultimate analysis of coal, coke: manufacture of coke by Otto – Hoffmann’s by-product coke oven process; Refining of Petroleum, Knocking-octane number of gasoline, cetane number of diesel oil. Synthetic Petrol; LPG , CNG.

Lubricants:- Definition, Mechanism of Lubrication, Properties of Lubricants (Definition and significance)

UNIT-III: Electrochemical cells and Corrosion:

Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion- material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electroless plating ; Paints.

UNIT-IV: Water technology:

Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods.

UNIT-V: Chemistry of Engineering Materials& Advanced Engineering materials

Cement:- Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete.

Refractories: - Definition, Characteristics, classification, Properties and failure of refractories.

Solar Energy: - Construction and working of Photovoltaic cell, applications.

Solid State Materials: Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors.

Liquid Crystals: - Definition – Classification with examples – Applications

Text Books:

1. Engineering Chemistry by Jain and Jain, Dhanpat Rai publishing co.
2. Engineering Chemistry by Willy India Pvt Ltd.
3. Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy; Silicon Publications.

Reference Books:

1. Engineering Chemistry by Shikha Agarwal; Cambridge University Press,2015 edition.
2. A text of Engineering Chemistry by S.S.Dara; S.Chand & Co Ltd.

ENGINEERING MECHANICS
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To impart the basic concepts of Engineering Mechanics and the principles of various force systems under static and dynamic conditions.
2. Comprehensive, theory based understanding of the underpinning natural and physical sciences and the engineering fundamentals applicable to the engineering discipline.
3. Application of established engineering methods to complex engineering problem solving.
4. The course is the key prerequisite course to sequences of courses dealing with mechanics of machines, stress analysis and design of mechanical systems.

Course Outcomes:

Students will be able to

1. Determine the resultant of the given force systems.
2. Analyze force systems using equations of equilibrium.
3. Determine centroid, center of gravity and moment of inertia of areas and bodies.
4. Analyze trusses and simple beams.
5. Distinguish between kinematics and kinetics.
6. Apply the work energy and impulse momentum methods of various engineering problems.

SYLLABUS

UNIT-I

Basic Concepts:

Scalar and vector quantities- Representation of vectors- Free vector force, Specification of force- Effect of force on rigid body- Free body diagram. Concurrent Forces in a plane: Principles of statics-Resolution and Composition of forces in a plane-Equilibrium of concurrent forces in a plane- Method of projections- Equilibrium of three forces in a plane Method of moments. Parallel Force system in a plane.

UNIT-II

Centroid & Moment of Inertia: Centroid & M.I – Area & Mass M.I – Radius of Gyration, Parallel axis– Perpendicular axis theorem – Simple Problems.

UNIT-III

General Case of Forces in a Plane: Resultant and equilibrium of general case of forces in a plane, Statically determinate plane trusses-Method of joints and Method of sections.

Friction – Coulombs laws of dry friction – Limiting friction, Problems on Wedge friction, Belt Friction-problems.

UNIT-IV

Dynamics of Particles - Rectilinear Motion – Kinematics, D'Alembert's principle, Kinetics – Work & Energy – Impulse Moment, Direct Central Impact – coefficient of restitution.

Curvilinear Motion – Kinematics, Projectile Motion, Moment of momentum, Work & Energy in Curvilinear motion.

UNIT-V

Dynamics of Rigid Bodies - Rigid body rotation – Kinematics - Kinetics – Work & Energy in Rigid body rotation, Plane Motion – Kinematics – Instantaneous center of rotation, Kinetics - Work-Energy principle in plane motion.

Text Book:

1. Engineering Mechanics by S.Timoshenko and D.HYoung McGraw-Hill.

Reference Books:

1. Engineering Mechanics, Vol.1 & 2 by J.L. Meriems and L.G. Kraige.
2. Engineering Mechanics by Singer.
3. Engineering Mechanics by K.L. Kumar, Tata Mc-Graw Hill.
4. Engineering mechanics by Bhavikatti, New age international.

ENGINEERING DRAWING
(Common to CIV, EEE & ME)

Lecture	: 1 Period	Int.Marks	: 30
Practice	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

Course Outcomes:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent the object in 3D view through isometric views.
5. Convert the isometric view to orthographic view and vice versa.

SYLLABUS

UNIT I

Polygons: Constructing regular polygons by general methods, inscribing and describing polygons on circles.

Curves: Parabola, Ellipse and Hyperbola by general methods, cycloids, involutes, tangents & normals for the curves.

UNIT II

Orthographic Projections: Horizontal plane, vertical plane, profile plane, importance of reference lines, projections of points in various quadrants, projections of lines, lines parallel either to one of the reference planes (HP,VP or PP)

Projections of straight lines inclined to both the planes, determination of true lengths, angle of inclination and traces- HT, VT

UNIT III

Projections of planes: regular planes perpendicular/parallel to one plane and inclined to the other reference plane; inclined to both the reference planes.

UNIT IV

Projections of Solids – Prisms, Pyramids, Cones and Cylinders with the axis inclined to one of the planes.

UNIT V

Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Text Books:

1. Engineering Drawing by N.D. Butt, Chariot Publications.
2. Engineering Drawing by Agarwal & Agarwal, Tata McGraw Hill Publishers

Reference Books:

1. Engineering Drawing by K.L.Narayana & P. Kannaiah, Scitech Publishers.
2. Engineering Graphics for Degree by K.C. John, PHI Publishers.
3. Engineering Graphics by PI Varghese, McGrawHill Publishers.
4. Engineering Drawing + AutoCad – K Venugopal, V. Prabhu Raja, New Age

ENVIRONMENTAL STUDIES
(Common to all Branches)

Lecture : 2 Periods

Tutorial : 1 Period.

Exam : 3 Hrs.

Int.Marks : 30

Ext. Marks : 70

Credits : 2

Course Objectives:

1. Developing an awareness and sensitivity to the total environment and its related problems.
2. Motivating students for active participation in environmental protection and improvement.
3. Developing skills for active identification and development of solutions to environmental problems
4. Evaluation of environment programmes in terms of social, economic, ecological and aesthetic factors.
5. Creating a “CONCERN AND RESPECT FOR THE ENVIRONMENT”.

Course Outcomes:

1. To bring awareness among the students about the nature and natural ecosystems
2. Sustainable utilization of natural resources like water, land, energy and air
3. Resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation
4. Safe guard against industrial accidents particularly nuclear accidents
5. Constitutional provisions for the protection of natural resources

SYLLABUS

UNIT – I

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness. Global Environmental Crisis

Ecosystems:

Basic Concepts - Structure and Functions of Ecosystems: Producers, Consumers and Decomposers. Types of Ecosystems: Forest Ecosystems, Grassland Ecosystems Desert Ecosystems and Aquatic Ecosystems

UNIT-II

Biodiversity:

Introduction to Biodiversity, Values of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Hotspots of Biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of Biodiversity.

UNIT-III

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture. **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people. **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems. **Energy Resources:** Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

UNIT-IV

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and Global warming, Acid rain and Ozone layer depletion. Solid Waste Management: Composting, Vermiculture, Urban and Industrial Wastes, Recycling and Reuse.

Environmental Problems in India:

Drinking water, Sanitation and Public health, Population growth and Environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

UNIT-V

Institutions and Governance:

Regulations by Government- Environmental Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act. Environmental Impact Assessment (EIA)

Case Studies:

Chipko Movement, Narmada Bachao Andolan, Silent Valley Project, Mathura Refinery and Taj Mahal, Industrialization of Patancheru, Nuclear reactor at Nagarjuna Sagar, Tehri Dam, Ralegaon Siddhi (Anna Hazare), Kolleru lake – Aquaculture, Fluorosis in Andhra Pradesh & Telangana.

Field Work:

Visit to a local area to document and mapping environmental assets. Visits to Industries, Water Treatment Plants, Effluent Treatment Plants.

Text Books:

1. Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada
2. Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford University Press.
3. Environmental Studies, Dr. M. Sita Rama Reddy, Dr. K. Anji Reddy, Silicon Publications, ASR Nagar, Bhimavaram
4. Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula Rani; Pearson Education, Chennai

Reference Books:

1. Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2. A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3. Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4. Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

**ENGINEERING CHEMISTRY LAB
(Common to CIV, EEE & ME)****Lab : 3 Periods**
Exam : 3 Hrs.**Int.Marks : 50**
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To investigate and understand Physical behavior in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world.
2. Effectively communicate experimental results and solutions to application problems through oral and written reports.
3. Understand the basic concepts, definitions, characteristics and phenomena.
4. Recognize the classical ideas and chemical phenomena and also define and analyse the concepts.

Course Outcomes:

1. An understanding of Professional and develop confidence on recent trends.
2. Able to gain technical knowledge of measuring, operating and testing of chemical instruments and equipments.
3. Acquire ability to apply knowledge of chemistry.
4. Exposed to the real time working environment.
5. Demonstrate the ability to learn Principles, design and conduct experiments.
6. Ability to work on laboratory and multidisciplinary tasks.

LIST OF EXPERIMENTS**Introduction to chemistry Laboratory**

1. Estimation of HCl using standard Sodium Hydroxide.
2. Determination of total hardness of water by EDTA method.
3. Estimation of Ferrous Iron by KMnO_4 .
4. Estimation of oxalic acid by KMnO_4
5. Estimation of Mohr's salt by $\text{K}_2\text{Cr}_2\text{O}_7$
6. Estimation of Dissolved oxygen by Winkler's method.
7. Determination of pH by pH meter and universal indicator method.
8. Conductometric titration of strong acid Vs strong base
9. Conductometric titration of strong acid Vs weak base.
10. Potentiometric titration of strong acid Vs strong base
11. Potentiometric titration of strong acid Vs weak base
12. Preparation of Phenol formaldehyde resin.
13. Determination of saponification value of oils
14. Determination of pour and cloud points of lubricating oil.
15. Determination Acid value of oil.

Text Books:

1. Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.Engineering College.

Reference Books:

1. Laboratory manual on Engineering Chemistry by Dr.Sudha Rani ;Dhanpat Rai Publishing Company.
2. Engineering Chemistry Laboratory manual – I & II by Dr. K.Anji Reddy; Tulip Publications.

ENGLISH COMMUNICATION SKILLS LAB- I
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn through practice the communication skills of listening, speaking, reading and writing.
2. To make students recognize the sounds of English through Audio- Visual aids.
3. To familiarize the students with stress and intonation.
4. To help the students build their confidence in speaking skills.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students improve their speaking skills in real contexts.
3. Students learn standard pronunciation and practice it daily discourse.
4. Students give up their communicative barriers.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. **'INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

**ENGINEERING WORKSHOP & IT WORKSHOP
(Common to CIV, EEE & ME)**

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

PART-A ENGINEERING WORKSHOP

Course Objective:

To impart hands-on practice on basic engineering trades and skills.

Course Outcomes:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Prepare jobs of various shapes using black smithy.
3. Make basic house wire connections.
4. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry	Fitting
1. T-Lap Joint 2. Cross Lap Joint 3. Dovetail Joint 4. Mortise and Tenon Joint	1. Vee Fit 2. Square Fit 3. Half Round Fit 4. Dovetail Fit
Black Smithy	Tin Smithy
1. Round rod to Square 2. S-Hook 3. Round Rod to Flat Ring 4. Round Rod to Square headed bolt	1. Taper Tray 2. Square Box without lid 3. Open Scoop 4. Funnel
House Wiring	
1. Parallel / Series Connection of three bulbs 2. Stair Case wiring 3. Florescent Lamp Fitting 4. Measurement of Earth Resistance	

Note: At least two exercises to be done from each trade.

Reference:

1. Elements of workshop technology, Vol.1 by S. K. and H. K. Choudary.

PART B: IT WORKSHOP:

Course Objectives:

1. Understand the basic components and peripherals of a computer.
2. To become familiar in configuring a system.
3. Learn the usage of productivity tools.
4. Acquire knowledge about the netiquette and cyber hygiene.
5. Get hands on experience in trouble shooting a system

Course Outcomes:

At the end of the course the students can able to

1. Assemble and disassemble the systems
2. Use the Microsoft office tools
3. Install various software
4. Know about various search engines
5. Trouble shoot various Hardware and Software problems
6. Use MATLAB and LATEX softwares

LIST OF EXERCISES

1. System Assembling, Disassembling and identification of Parts / Peripherals
2. Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.
3. MS-Office / Open Office
 - a) Word - Formatting, Page Borders, Reviewing, Equations, symbols.
 - b) Spread Sheet - organize data, usage of formula, graphs, charts.
 - c) Power point - features of power point, guidelines for preparing an effective presentation.
 - d) Access- creation of database, validate data.
4. Network Configuration & Software Installation-Configuring TCP/IP, proxy and firewall settings. Installing application software, system software & tools.
5. Internet and World Wide Web-Search Engines, Types of search engines, netiquette, cyber hygiene.
6. Trouble Shooting-Hardware trouble shooting, Software trouble shooting.
7. MATLAB- basic commands, subroutines, graph plotting.
8. LATEX-basic formatting, handling equations and images.

Reference Books:

1. Computer Hardware, Installation, Interfacing, Troubleshooting and Maintenance, K.L. James, Eastern Economy Edition.
2. Microsoft Office 2007: Introductory Concepts and Techniques, Windows XP Edition By Gary B. Shelly, Misty E. Vermaat and Thomas J. Cashman (2007, Paperback).
3. LATEX- User's Guide and Reference manual, Leslie Lamport, Pearson, LPE, 2/e.
4. Getting Started with MATLAB: A Quick Introduction for Scientists and engineers, Rudraprathap, Oxford University Press, 2002.
5. Scott Mueller's Upgrading and Repairing PCs, 18/e, Scott. Mueller, QUE, Pearson, 2008
6. The Complete Computer upgrade and repair book, 3/e, Cheryl A Schmidt, Dreamtech.
7. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dreamtech.
8. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.

INNER ENGINEERING
(Common to CIV, EEE & ME)

Practice : 2 Periods

Course Objectives:

Student should learn:

1. Human values and tools to lead a happy, stress-free life.
2. Yoga asanas, Pranayama, Sudarshan Kriya & Meditation
3. At-least two creative arts out of photography, sketching, craft-making, singing, clay molding, upcycling etc.
4. Concentration pranayama, Ego bursting process.
5. To take up responsibility for society and teach classes of their choice to school children.
6. About good food habits for good health.

Course Outcomes:

Student should be capable of

1. To improve his concentration levels and improve his public speaking abilities.
2. To balance his academic and non-academic activities (Work Life Balance).
3. To widen his vision and increase his breadth of perspective in his journey of 4 years.
4. To improve his communications skills, leadership, teamwork and decision-making abilities.
5. To inculcate creativity & innovation, planning & organizing as part of their life.
6. Taking responsibility for themselves and people around them.
7. To make their journey more fun and enjoyable.

SYLLABUS

UNIT-I

YES!+ Workshop:

Yoga Postures – Seven Layers To our Existence – Puzzles – Sources Of Energy – Live in the present Moment – Importance of Breath – Ujjai Breath – Pranayama – Sudarshana Kriya.

UNIT-II

YES!+ Workshop:

Yoga Postures (Suryanamaskars) – Giving 100% in everything – Time management – Happiness point – Opposite Values – Pranayama – Sudarshan kriya

UNIT-III

YES!+ Workshop:

Yoga Postures – Knowledge points (Acceptance, opinions discretion and handling mistakes) – Eye Gazing Process – Dance – Life Story process – Sudarshana Kriya (short) – Eternal life – Ego Bursting – Relationships – Parents – Studies – Compliments/Praising process.

UNIT-IV

Creative Arts:

Photography – Sketching – Handy-crafts – Clay molding – Singing – Upcycling – Communing with nature – Creative writing.

UNIT -V

Service:

Leadership in action – Contributing to society – Take up Responsibility –Empowerment – Public Speaking – Art of Teaching.

Reference Books:

1. Discourse on Patanjali Yoga Sutras by H.H Sri SriRavishankar
2. Human values and professional ethics byRRGaur,RSangal,GPBagaria
3. The Art of Photography by AL Judge
4. Hand made in India : Crafts of India by Ranjan Aditi

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R17)
I/IV B.TECH
 (With effect from **2017-2018** Admitted Batch onwards)
 Under Choice Based Credit System

GROUP-A (CIVIL, EEE & MECHANICAL)
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1201	English – II *	3	3	1	--	4	30	70	100
B17 BS 1202	Mathematics – II	3	3	1	--	4	30	70	100
B17 BS 1203	Mathematics – III*	3	3	1	--	4	30	70	100
B17 BS 1204	Engineering Physics	3	3	1	--	4	30	70	100
B17 CS 1201	Computer Programming Using C	3	3	1	--	4	30	70	100
# DS 1	Department Subject	3	3	1	--	4	30	70	100
B17 BS 1206	Engineering Physics Lab	2	--	--	3	3	50	50	100
B17 BS 1208	English Communication Skills Lab – II *	2	--	--	3	3	50	50	100
B17 CS 1204	C Programming Lab	2	--	--	3	3	50	50	100
B17 BS 1210	Engineering Physics Virtual Labs- Assignments	--	--	--	2	2	--	--	--
B17 BS 1211	NCC	--	--	--	2	2	--	--	--
Total		24	18	6	13	37	330	570	900

*** Common to both Group - A and Group - B**

#DS 1	CIVIL	B17 CE 1201	Building Materials and Construction
	EEE	B17 EE 1201	Circuit Theory
	MECHANICAL	B17 EE 1202	Basic Electrical and Electronics Engineering

ENGLISH - II
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop spoken and written forms of the students in both formal and informal situations.
4. To enrich the vocabulary of the students through the usage of the lexis in appropriate contexts.
5. To expose them to the correct structural patterns of English.

Course Outcomes: At the end of the Course, Student will be able to:

1. To comprehend the speech of people belonging to different backgrounds and regions.
2. Understand the importance of speaking and writing for personal and professional communication and practice it in real contexts.
3. To express fluently and accurately in social discourse.
4. Participate in group activities like role-plays, discussions and debates.
5. Identify the discourse features, and improve intensive and extensive reading skills.

SYLLABUS

UNIT I:

- A. Detailed-Text: Unit 1: ' The Greatest Resource- Education'
- B. Non-Detailed Text: Lesson 1: ' A P J Abdul Kalam' from The Great Indian Scientists.

UNIT II:

- A. Detailed-Text: Unit 2: ' A Dilemma'
- B. Non-Detailed Text: Lesson 2:'C V Raman' from The Great Indian Scientists.

UNIT III:

- A. Detailed-Text: Unit 3: 'Cultural Shock': Adjustments to new Cultural Environments
- B. Non-Detailed Text: Lesson 3:'Homi Jehangir Bhabha' from The Great Indian Scientists.

UNIT IV:

- A. Detailed-Text: Unit 4: 'The Lottery'
- B. Non-Detailed Text: Lesson 4: 'Jagadish Chandra Bose' from The Great Indian Scientists.

UNIT V:

- A. Detailed-Text: Unit 5: ' The Chief Software Architect'
- B. Non-Detailed Text: Lesson 5: ' Prafulla Chandra Ray' from The Great Indian Scientists

Detailed Textbook:

1. English Encounters Published By Maruthi Publishers.

Non-Detailed Text Book:

1. The Great Indian Scientists Published by Cengage learning.

MATHEMATICS – II
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand some basic numerical methods to solve algebraic and transcendental equations.
4. Learn interpolation methods for equally spaced and unequally spaced data points. Learn methods for numerical evaluation of integrals and for solving first order ODEs.
5. Learn concepts of a Fourier series and Fourier transform.

Course Outcomes: At the end of the Course, Student will be able to:

1. Find a real root of algebraic and transcendental equations using different methods.
2. Know the relation between the finite difference operators. Determine interpolation polynomial for a given data.
3. Evaluate numerically certain definite integrals applying Trapezoidal and Simpson's rules.
4. Solve a first order ordinary differential equation by Euler and RK methods.
5. Find Fourier series of a given function satisfying Dirichlet conditions. Find half range cosine and sine series for appropriate functions.
6. Find Fourier transforms Fourier cosine and sine transforms of appropriate functions and evaluate certain integrals using inverse transforms and Fourier integral.

SYLLABUS

UNIT I: Solution of Algebraic and Transcendental Equations:

Introduction, Bisection method, Method of false position, Iteration method, Newton- Raphson method (One variable and simultaneous Equations).

UNIT II: Interpolation:

Introduction, Errors in polynomial interpolation, Finite differences, Forward differences, Backward differences, Central differences and Symbolic relations between the operators, Differences of a polynomial, Newton's formulae for interpolation, Interpolation with unequal intervals, Lagrange's interpolation formula.

UNIT III: Numerical Integration and solution of Ordinary Differential equations:

Trapezoidal rule, Simpson's $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rules, Solution of ordinary differential equations by Taylor series method, Picard's method of successive approximations, Euler's method, Runge-Kutta methods (second order and fourth order).

UNIT IV: Fourier Series:

Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Even and odd functions, Change of interval, Half-range sine and cosine series, Parseval's formula.

UNIT V: Fourier Transforms:

Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, properties, inverse transforms, Parseval's identities, Finite Fourier transforms.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
2. V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
4. David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press.
5. Srimanta Pal, Subodh C. Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

MATHEMATICS - III
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand important concepts of Linear algebra including solution of linear simultaneous equations, eigen values and eigen vectors.
4. Learn Beta, Gamma functions and how to evaluate double and triple integrals.
5. Learn fundamental concepts of vector calculus that help understand other Engineering courses.

Course Outcomes: At the end of the Course, student will be able to:

1. Determine rank, and solve a system of linear simultaneous equations numerically using various matrix methods.
2. Determine Eigen values and Eigen vectors of a given matrix, Reduce a Quadratic form to its canonical form and classify.
3. Evaluate double integrals over a region and triple integral over a volume.
4. Use the knowledge of Beta and Gamma functions in evaluation of different integrals.
5. Find gradient of a scalar function, divergence and curl of a vector function. Use vector identities for solving problems.
6. Evaluate line, surface and volume integrals by the use of Green's, Stokes' and Gauss divergence theorems.

SYLLABUS

UNIT I: Linear systems of equations:

Rank, Echelon form, Normal form, Solution of linear systems, Gauss elimination, Gauss-Jordan, Jacobi and Gauss-Seidel methods.

Applications: Finding the current in electrical circuits.

UNIT II: Eigen values - Eigen vectors and Quadratic forms:

Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix by using Cayley-Hamilton theorem, Diagonalization, Quadratic forms, Reduction of a Quadratic form to Canonical form, Rank, Positive, Negative, Semi-Definite and indefinite forms of a Quadratic form, Index and Signature of a Quadratic form.

Applications: Free vibration of a two-mass system.

UNIT III: Multiple integrals:

Double and triple integrals, Change of variables, Change of order of integration. Application to finding Areas, Moment of Inertia and Volumes.

Beta and Gamma functions, Properties, Relation between Beta and Gamma functions, Application to evaluation of improper integrals. The error function and the complimentary error function.

UNIT IV: Vector Differentiation:

Gradient, directional derivative, Divergence, Curl, Incompressible flow, solenoidal and irrotational vector fields, second order operators, vector identities.

UNIT V: Vector Integration:

Line integral, Work done, Potential function; Area, Surface and volume integrals, Flux, Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof) and related problems.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson edn.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
3. Peter O'Neil, Advanced Engineering Mathematics, 7th edition, Cengage Learning.
4. D.W. Jordan and T. Smith, Mathematical Techniques, Oxford University Press.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING PHYSICS
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

To re-orient Physics curriculum to the needs of the circuit and non circuit branches of Engineering / Technology courses offered by SRKREC(A) Bhimavaram that serves to understand the general and branch specific basic topics and introduce the related advanced technologies to the stake holders.

1. To impart the knowledge of physical optics phenomena like Interference and Diffraction required to understand the design and working of optical instruments with high resolution.
2. To understand the concept of coherence and generation of high intensity light sources and propagation of light waves in fibers for technological applications.
3. To study and analyze the behavior of electric and magnetic fields and their mutual interaction, and to understand the science of ultrasonics.
4. To understand the basic concepts of Quantum mechanics to know the behavior of fundamental particles in materials and to understand the basic nature of materials in general and classify them.
5. To study the structure property relationship of crystalline solids and understand basics of nanotechnology.

Course Outcomes: At the end of the course the students-

1. Learn the basic concepts of interference and diffraction of light and its applications.
2. Understand the science of producing high intensity light beams for technological applications and also understand the propagation of light waves in optical fibers in various applications.
3. Understand the inter relationship of electric and magnetic fields and learn ultrasonic's as a tool for technological applications
4. Learn the behavior of particles at the very microscopic level by using wave nature of particles and understand the behavior of materials and be able to classify them using the band theory of solids.
5. Learn the basics of structures of solid materials and nano material preparation Techniques/methods.

SYLLABUS

UNIT I: Interference and Diffraction

Principle of superposition-coherence-interference in thin films (reflected system) – Wedge shaped film-Newton's rings-Michelson's interferometer. Fraunhofer's diffraction at single slit, Diffraction grating-Resolving power of a grating.

UNIT- II: Lasers and Optical Fibers

Introduction, Spontaneous emission and Stimulated emission – Einstein's relation – Requirements of Laser device- Ruby laser- He-Ne gas laser- Characteristics of laser- Applications.

Description of optical fiber, Principle of light propagation- Optical fiber –Acceptance angle-Numerical aperture of optical fiber- Modes of propagation- Classification of fibers- Applications of fiber.

UNIT- III: Electro Magnetic Fields and Ultrasonics

Concept of Electromagnetic induction , Faraday's law, Lenz's law, Electric fields due to time varying magnetic fields, Magnetic fields due to time varying electric fields, Displacement current, Modified Ampere's law, Maxwell's equations and their significance (without derivation).

Definition of Ultrasonics-Methods of Producing Ultrasonics- Detection of Ultrasonics- Applications of Ultrasonics.

UNIT- IV: Quantum Mechanics and Band Theory of Solids

Introduction, de Broglie matter waves- properties-Experimental confirmation, wave function-significance- Schrodinger's time dependent and time independent wave equations- Eigen values and functions, Particle in a box.

Band theory of Solids- Introduction- Kronig Penney model (Qualitative)- Energy bands of crystalline solids- Distinction between Conductors, Semi conductors and insulators.

UNIT-V: Crystallography and Nano Materials

Basis and Lattice, Crystal systems, Bravais lattice, Unit cell Coordination number – Packing fraction for SC ,FCC, and BCC lattices, Miller indices- Diffraction of X rays from crystals- Bragg's law.

Introduction to Nanomaterials – Synthesis methods : Condensation, ball milling, sol-gel, chemical vapour deposition methods, properties and applications.

Text Books:

1. Physics by Resnick & Halliday. Wiley - Eastern (India) Ltd.
2. Engineering Physics By M.N. Avadhanulu & PG Kshirasagar, S.Chand & Co.
3. Engineering Physics by V. Rajendran, Mc Graw Hill Education (India) Pvt.Ltd.

Reference Books:

1. Engineering Physics by MR Srinivasan, New International Publishers.
2. Solid State Physics – A.J. Dekkar, MacMillan (India) Ltd.
3. Engineering Physics by Gaur Gupta, Dhanpat Rai Publications, Meerut, India.
4. Engineering Physics by P k Palanisamy, Scitech Publications (India) Pvt.Ltd

(Note: Assignment Marks of **Engineering Physics** are to be considered from the Internal marks of **Engineering Physics-- Virtual Labs – Assignments B17 BS 1210**)

COMPUTER PROGRAMMING USING C
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Formulating algorithmic solutions to problems and implementing algorithms in C

1. Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux
2. understanding branching, iteration and data representation using arrays
3. Modular programming and recursive solution formulation
4. Understanding pointers and dynamic memory allocation
5. Understanding miscellaneous aspects of C
6. Comprehension of file operations

Course Outcomes:

1. Understand the basic terminology used in computer programming
2. Write, compile and debug programs in C language.
3. Use different data types in a computer program.
4. Design programs involving decision structures, loops and functions.
5. Explain the difference between call by value and call by reference
6. Understand the dynamics of memory by the use of pointers
7. Use different data structures and create/update basic data files.

SYLLABUS

UNIT I:

Unit objective: Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux

Introduction: Computer systems, Hardware and Software Concepts.

Problem Solving: Algorithm / Pseudo code, flowchart, program development steps, computer languages: machine, symbolic and highlevel languages, Creating and Running Programs: Writing, Editing(vi/emacs editor), Compiling(gcc), Linking and Executing in under Linux.

BASICS OF C: Structure of a c program, identifiers, basic data types and sizes. Constants, Variables, Arithmetic , relational and logical operators, increment and decrement operators, conditional operator, assignment operator, expressions, type conversions, Conditional Expressions, precedence and order of evaluation, Sample Programs.

UNIT II:

Unit objective: understanding branching, iteration and data representation using arrays

SELECTION – MAKING DECISION: TWO WAY SELECTION: if-else, null else, nested if, examples, Multi-way selection: switch, else-if, examples.

ITERATIVE: loops- while, do-while and for statements , break, continue, initialization and updating, event and counter controlled loops, Looping applications: Summation, powers, smallest and largest.

ARRAYS: Arrays- concepts, declaration, definition, accessing elements, storing elements, Strings and String Manipulations, 1-D arrays, 2-D arrays and character arrays, string manipulations, Multidimensional arrays, array applications: Matrix operations, checking the symmetricity of a Matrix. **STRINGS: concepts, c strings.**

UNIT III:

Objective: Modular programming and recursive solution formulation

FUNCTIONS- MODULAR PROGRAMMING: functions, basics, parameter passing, storage classes extern, auto, register, static, scope rules, block structure, user defined functions, standard library functions, recursive functions, Recursive solutions for fibonacci series, towers of Hanoi, header files, C Preprocessor, example c programs, Passing 1-D arrays, 2-D arrays to functions.

UNIT IV:

Objective: Understanding pointers and dynamic memory allocation

POINTERS: pointers- concepts, initialization of pointer variables, pointers and function arguments, passing by address- dangling memory, address arithmetic, character pointers and functions, pointers to pointers, pointers and multi-dimensional arrays, dynamic memory management functions, command line arguments

UNIT V:

Objective: Understanding miscellaneous aspects of C

ENUMERATED, STRUCTURE AND UNION TYPES: Derived types- structures declaration, definition and initialization of structures, accessing structures, nested structures, arrays of structures, structures and functions, pointers to structures, self referential structures, unions, typedef, bit-fields, program applications

BIT-WISE OPERATORS: logical, shift, rotation, masks.

Objective: Comprehension of file operations

FILEHANDLING: Input and output- concept of a file, text files and binary files, Formatted I/O, File I/O operations, example programs

Text Books:

1. Problem Solving and Program Design in C, Hanly, Koffman, 7thed, PERSON
2. Programming in C, Second Edition PradipDey and Manas Ghosh, OXFORD Higher Education
3. Programming in C, A practical approach Ajay Mittal PEARSON
4. The C programming Language by Dennis Richie and Brian Kernighan
5. Programming in C, B. L. Juneja, Anith Seth, Cengage Learning.

Reference Books:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming with C, Bichkar, Universities Press
3. Programming in C, ReemaThareja, OXFORD
4. C by Example, Noel Kalicharan, Cambridge

BUILDING MATERIALS AND CONSTRUCTION
(For CIVIL)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To promote development, production, standardization and large-scale application of cost-effective innovative building materials and construction technologies in housing and building sector.
2. To develop and promote methodologies and technologies related to different Building constructions
3. To promote new building materials and components through technical support and encouraging entrepreneurs to set up production units.
4. To provide standard scientific and technical services to professionals, construction agencies and entrepreneurs in selection, evaluation and design engineering materials.

Course Outcomes:

1. Define and classify various stones, clay products used in construction sector.
2. Find various types of wood their conversion and relevant BIS testing procedures to be carried out to ascertain the quality of building materials.
3. Identify the major ingredients of construction materials like lime, cement, aggregate and their use in the construction industry.
4. Select different materials for finishing's and various constructions pertaining to masonry works and foundations.
5. Develop the conceptual knowledge of various supports in building construction.

SYLLABUS

UNIT I: STONES, BRICKS AND CLAY PRODUCTS

Stones: Classification of stones, Properties of building stones, Stone quarrying, precautions in blasting **Bricks:** Classification of Bricks, Manufacture of Bricks, general qualities of Bricks as per IS code, tests for good bricks as per IS code, including field tests. **Clay Products:** Tiles-types, manufacturing and their uses, Earth-ware, Terra-cotta, stone ware, Porcelain.

UNIT II: WOOD, WOOD BASED PRODUCTS

Wood: cross section details of trees, their general properties, characteristics of good timber defects in timber, mechanical properties of timber, seasoning and its importance, Decay of timber, **Wood based Products:** Veneers, Plywood and its types, Manufacturing of plywood, plywood grades as per IS code, Laminated wood, merits of plywood and laminated wood, Lamin Boards, Block boards, Batten board, Particle boards

UNIT III: LIME, CEMENT & AGGREGATES

Lime: Various ingredients of lime, Constituents of lime stone, classification of lime ,
Cement: Natural and artificial cements, types of artificial cements and their uses, Wet and dry process of manufacturing ordinary Portland cement (OPC), composition of cement, Various field and Laboratory tests on OPC as per IS code, Storage of cement. **Aggregates:** Classification of aggregate – Coarse and fine aggregates, Particle shape and Texture, Specific gravity, Bulk density, porosity and Absorption, Moisture content of Aggregate – Bulking of sand, Sieve analysis.

UNIT IV: FINISHINGS, MASONRY AND FOUNDATIONS

Finishings: Paints and Varnishes: Constituents and characteristics of paints, types of paint and their uses, painting defects, causes and remedies. Constituents of varnishes, types of varnish and their uses, Pointing and Plastering. **Masonry:** Different types of Stone Masonry- Plan, Elevation, Sections of stone Masonry works- Brick Masonry- Different Types of Bonds- Plan, Elevation and section of Brick Bonds upto Two-Brick wall thickness- Partition walls- Different types of Block Masonry- Hollow concrete Blocks- FAL-G Blocks, Hollow Clay Blocks. **Foundations:** Types- strip, isolated, strap, combined footings, Raft-Mat- flat slab and Beam raft, box type raft.

UNIT V: ROOFING, FORM WORK & SCAFFOLDING

Roofing: Mangalore tiled roof, RCC roof, Madras terrace roof, Hollow tiled roof, Asbestos cement, Fibre glass, Aluminum G.I. Sheet roofing's. **Form work, Scaffolding:** form work- types of formwork, centering- scaffolding- types of scaffolding. Trusses: Types- King post and queen post trusses and their uses. Stair cases: Various types of stair cases- dog legged, quarter landing, spiral stairs etc.

Text Books:

1. Engineering Materials by Rangawala. Charotar Publications
2. Building Materials by S.K.Duggal, New Age International Publications.
3. Building Materials by B.C. Punmia, Laxmi Publications private ltd.
4. Building Construction by B.C. Punmia, Laxmi Publications (p) ltd.

Reference Books:

1. Building Materials by S.S. Bhavikatti, Vices publications House private ltd.
2. Building Construction by S.S. Bhavikatti, Vices publications House private ltd.
3. Building Materials by M.L.Gambhir, Tata McGraw Hill Publishing Co. Ltd. New Delhi.
4. Building Materials by P.C.Vergheze, PHI learning (P) ltd.
5. Building construction by P.C.Vergheze, PHI Learning (P) Ltd.

CIRCUIT THEORY (For EEE)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study the concepts of passive elements, types of sources and various network reduction techniques.
2. To understand the applications of network topology to electrical circuits.
3. To study the concept of magnetic coupled circuits.
4. To understand the behavior of RLC networks for sinusoidal excitations and to understand the concept of resonance.
5. To understand the concepts of 3-phase circuits.

Course Outcomes:

Students are able to solve

1. Various electrical networks in presence of active and passive elements.
2. Electrical networks with network topology concepts.
3. Magnetic circuit with various dot conventions.
4. R, L, C network with sinusoidal excitation.
5. Three phase AC circuits.

SYLLABUS

UNIT-I: Introduction to Electrical Circuits:

Passive components and their V-I relations. Sources (dependent and independent) -Kirchhoff's laws, Network reduction techniques (series, parallel, series - parallel, star-to-delta and delta-to-star transformation). Source transformation technique, nodal analysis and mesh analysis.

UNIT-II: Network topology:

Definitions of Graph and Tree, Basic cutset and tieset matrices for planar networks, Loop and nodal methods of analysis of networks with dependent and independent voltage and current sources, Duality and Dual networks.

UNIT-III: Magnetic Circuits:

Basic definition of MMF, flux and reluctance. Analogy between electrical and magnetic circuits. Faraday's laws of electromagnetic induction Concept of self and mutual inductance. Dot convention-coefficient of coupling and composite magnetic circuit. Analysis of series and parallel magnetic circuits.

UNIT-IV: Single Phase A.C Systems:

Periodic waveforms (determination of rms, average value and form factor). Concept of phase angle and phase difference – Waveforms and phasor diagrams for lagging, leading networks. Complex and polar forms of representations, steady state analysis of R, L and C circuits. Power Factor and its significance - Real, Reactive, Apparent and Complex power. Node and mesh analysis of AC networks, Series and parallel resonance. Numerical problems.

UNIT-V: Three Phase Circuits:

Advantages of Three Phase Circuits, Balanced and Unbalanced systems, Relation between Line and Phase Quantities in Star and delta connected circuits, Analysis of Balanced & Unbalanced Three Phase Circuits.

Text Books:

1. Engineering Circuit Analysis By W.H. Hayt Jr ,J.E. Kemmerly,S.M.Durbin 8th Ed.,Mc.Graw Hill.
2. Network Analysis: Van Valkenburg; Prentice-Hall of India Private Ltd
3. Basic Electrical Engineering by V.K Mehta & Rohit Mehta- 2006 Edition S.Chand & Company

Reference Books:

1. Circuit Theory Analysis and Synthesis., 7th Edition 2014 by Abhijit Chakrabarthi,Dhanpat Rai&Co.
2. Fundamentals of Electric circuits by Charles k Alexander & M N O Sadiku 5thEd., Mc.Graw Hill.
3. Circuits & Networks by Sudhakar & Syam Mohan, 4th edition, 2010 Mcgraw Hill.

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(For ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn the basic principles of electrical and magnetic circuits and analyze the networks
2. To understand the principle of operation and construction details of DC machines.
3. To understand the principle of operation and construction details of Transformers.
4. To understand the principle of operation and construction details of three phase Induction motor and Alternators.
5. To study the operation of PN junction diode, Half -wave Bridge rectifiers, PNP, NPN Transistors and various Amplifiers.

Course Outcomes:

1. Able to analyze the various Electrical networks and understand the basics of Magnetic Circuits
2. Able to understand the operation of DC generators,3-Point starter and conduct the Swinburne's test.
3. Able to analyze the Performance of Transformers.
4. Able to explain the operation of three phase induction motors and alternator.
5. Able to analyze the operation of Half-wave and Full-wave rectifiers and single stage CE amplifier.

SYLLABUS**UNIT I: Electrical and Magnetic Circuits:**

Basic definitions, Types of network elements, Ohm's Law, Kirchhoff's Laws, Series and parallel Circuits and star-delta and delta-star transformations-simple problems. Magnetic flux, MMF, Reluctance, Faraday's laws, Lenz's law, statically induced EMF, dynamically induced EMF.

UNIT-II: DC Machines:

Principle of operation of DC generator- EMF equation-Types of DC Generators-DC motor Types-Torque equation-Applications-Swinburne's Test, Speed control methods.

UNIT-III: Transformers:

Principle of operation of Single phase Transformers- EMF equation-losses-OC and SC Tests-Efficiency and Regulation.

UNIT-IV: AC Machines:

Principle of operation of Three phase Induction motor-Slip-Torque characteristics-Efficiency-applications- Principle of operation of Alternator-EMF equation, Regulation of alternator by synchronous Impedance method.

UNIT-V: Diodes-Rectifiers and Transistors:

PN junction diode-Forward bias and reverse bias operation, V-I characteristics-Diode applications (Half wave, Full wave and bridge rectifier), Zener diode.

PNP and NPN junction Transistors, Transistor as an amplifier, single stage CE amplifier, Frequency response of CE amplifier.

Text Books:

1. Electrical Technology by Surinder Pal Bali, Pearson Publications. January 1, 2013
2. Electronic Devices and Circuits, R.L Boylestad and Louis Nashelsky, 9th Edition, PEI/PHI 2006.
3. Principles of Electrical Engineering and Electronics, V.K mehta, Published December 1st 2006 by S. Chand & Company Ltd.
4. Principles of Electronics Engineering, V.K mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2012

Reference Books:

1. Basic Electrical Engineering, by J.B Gupta, K Kataria and Sons; Reprint 2013 edition.
2. Fundamentals of Electrical Engineering and Electronics, by Theraja B.L, S Chand Multicolor edition (1 December 2006)

**ENGINEERING PHYSICS LAB
(Common to CIV, EEE & ME)**

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

Training the field oriented Engineering graduates to handle instruments and their design methods to improve the accuracy of measurements.

1. To impart hands-on experience to the students entering engineering/Technology education about handling sophisticated equipment/ instruments.
2. To make the students understand the theoretical aspects of various phenomena experimentally.

Course Outcomes:

Physics lab curriculum gives fundamental understanding of design of an instrument with targeted accuracy for physical measurements.

1. Students get hands on experience in setting up experiments and using the instruments/equipment individually.
2. Get introduced to using new/ advanced technologies and understand their significance.

**LIST OF EXPERIMENTS
(Any 10 of the following listed experiments)**

1. Determination of wavelength of a source-Diffraction Grating-Normal incidence.
2. Newton's rings – Radius of Curvature of Plano - Convex Lens.
3. Determination of thickness of a spacer using wedge film and parallel interference fringes.
4. Determination of Rigidity modulus of a material- Torsional Pendulum.
5. Determination of Acceleration due to Gravity and Radius of Gyration-Compound Pendulum.
6. Melde's experiment – Transverse and Longitudinal modes.
7. Verification of laws of vibrations in stretched strings – Sonometer.
8. Determination of velocity of sound – Volume Resonator.
9. L- C- R Series Resonance Circuit.
10. Study of I/V Characteristics of Semiconductor diode.
11. I/V characteristics of Zener diode.
12. Characteristics of Thermistor – Temperature Coefficients.
13. Magnetic field along the axis of a current carrying coil – Stewart and Gee's apparatus.
14. Energy Band gap of a Semiconductor p - n junction.
15. Hall Effect in semiconductors.
16. Time constant of CR circuit.
17. Determination of wavelength of laser source using diffraction grating.
18. Determination of Young's modulus by method of single cantilever oscillations.
19. Determination of lattice constant – lattice dimensions kit.
20. Determination of Planck's constant using photocell.
21. Determination of surface tension of liquid by capillary rise method.

Reference Book:

1. Advanced Practical Physics –Vol 1 & 2 By S.Venkata Raman, S.Chand & Co.

ENGLISH COMMUNICATION SKILLS LAB- II
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn demonstratively the communication skills of listening, speaking, reading and writing.
2. To enable the students participate in group interactions.
3. To improve the presentation skills of the students.
4. To help the students gain their confidence in attending the interviews.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students enhance their presentation skills.
3. Students participate in group discussions and improve their team skills.
4. Students confidently face the interviews.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. **'INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

C PROGRAMMING LAB
(Common to CIV, EEE & ME)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

List of Programs

Exercise - 1 Basics

- a) What is an OS Command, Familiarization of Editors - vi, Emacs
- b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
- c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers
From Command line

Exercise - 2 Basic Math

- a) Write a C Program to Simulate 3 Laws at Motion
- b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

- a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
- b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

- a) Write a C Program to Find Whether the Given Number is
 - i) Prime Number
 - ii) Armstrong Number
- b) Write a C program to print Floyd Triangle
- c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

- a) Write a C Program demonstrating of parameter passing in Functions and returning values.
- b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III)

- a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch...case
- b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

Write a C Program to compute the values of $\sin x$ and $\cos x$ and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

- a) Search-Linear.
- b) Sorting-Bubble, Selection.
- c) Operations on Matrix.

Exercises - 9 Structures

- a) Write a C Program to Store Information of a Movie Using Structure
- b) Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
- c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

- a) Write a C Program to Access Elements of an Array Using Pointer
- b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

- a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.
- b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

Understand the difference between the above two programs

Exercise – 12 Strings

- a) Implementation of string manipulation operations **with** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare
- b) Implementation of string manipulation operations **without** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare

Exercise -13 Files

- a) Write a C programming code to open a file and to print its contents onscreen.
- b) Write a C program to copy files

Exercise - 14 Files Continued

- a) Write a C program merges two files and stores their contents in another file.
- b) Write a C program to delete a file.

Note:

- a) All the Programs must be executed in the Linux Environment. (Mandatory)
- b) The Lab record must be a print of the LATEX (.tex) Format.

ENGINEERING PHYSICS - VIRTUAL LABS – ASSIGNMENTS
(Common to CIV, EEE & ME)

Lab : 2 Periods

Int.Marks: 5

Course Objective: Training Engineering students to prepare a technical document and improving their writing skills.

Course Outcome: Physics Virtual laboratory curriculum in the form of assignment ensures an engineering graduate to prepare a /technical/mini-project/ experimental report with scientific temper.

List of Experiments

1. Hall Effect
2. Crystal Structure
3. Hysteresis
4. Brewster's angle
5. Magnetic Levitation / SQUID
6. Numerical Aperture of Optical fiber
7. Photoelectric Effect
8. Simple Harmonic Motion
9. Damped Harmonic Motion
10. LASER – Beam Divergence and Spot size
11. B-H curve
12. Michelson's interferometer
13. Black body radiation

URL: www.vlab.co.in

(Note: Internal Marks of **Engineering Physics - Virtual Labs – Assignments** are to be considered as Assignment marks in the Internal Marks of **Engineering Physics- B17 BS 1204**)

Code: B17 BS 1211

NCC
(Common to CIV, EEE & ME)

Practice : 2 Periods

The NCC- National Integration and Awareness- Drill- Personality Development Life Skills- Leadership- Disaster Management-Social Awareness and Community Development- Health and Hygiene- Environment Awareness and Conservation.

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

II/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

ELECTRICAL AND ELECTRONICS ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 2101	Mathematics - IV	3	3	1	--	4	30	70	100
B17 EC 2101	Electronics Devices & Circuits	3	3	1	--	4	30	70	100
B17 EE 2101	Network Analysis and Synthesis	3	3	1	--	4	30	70	100
B17 EE 2102	Electro Magnetic Field Theory	3	3	1	--	4	30	70	100
B17 EE 2103	Electrical Measurements & Instruments	3	3	1	--	4	30	70	100
B17 BS 2104	Engineering Economics	3	3	1	--	4	30	70	100
B17 EE 2105	Networks & Measurements Lab	2	--	--	3	3	50	50	100
B17 EC 2105	Electronics Devices & Circuits Lab	2	--	--	3	3	50	50	100
B17 BS 2106	Programming Skills-I	1	--	--	2	2	50	---	50
B17 BS 2107	English Proficiency-I	--	1	1	--	2	--	--	--
B17BS 2108	Professional Ethics & Human Values	--	2	--	--	2	--	--	--
Total		23	21	7	8	36	330	520	850

MATHEMATICS IV
(Common to CE,ECE,EEE& ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students should learn

1. The concept of Analytic function, its implications and applications in flow problems.
2. Solution of one- dimensional wave equation, one-dimensional heat equation and two-dimensional Laplace equation by the use of 'separation of variables'.
3. Formation and solution of linear difference equations. Important concepts of Z-transform and its use to solve linear difference equations.
4. Basic concepts of certain discrete and continuous probability distributions.
5. Concepts of Sampling theory for analyzing large and small samples.

Course Outcomes:

Students will be capable of

1. Using the concept of Analytic function in applications including Electrostatics and Fluid dynamics.
2. Finding theoretical solution of certain Elliptic, Parabolic and Hyperbolic partial differential equations.
3. Using Z-transforms to solve linear difference equations with constant coefficients.
4. Fitting of probability frequency distribution to a given data.
5. Using the concepts of sampling theory to analyze data related to some large and small samples.

SYLLABUS

UNIT-I Functions of a Complex Variable

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n , e^z , $\sin z$, $\sinh z$ and $\log z$ (**no questions may be set**).

Limit and continuity of a function of the complex variable, derivative, analytic function, entire function, Cauchy- Riemann equations, finding an analytic function, Milne-Thomson method, Applications of analytic function to flow problems, and in Electrostatics. Conformal mapping: the transformations defined by $w = z+c$, $w = cz$, $w = 1/z$, the Bilinear transformation, $w = z^2$ and $w=e^z$.

UNIT-II Applications of Partial Differential Equations

Method of separation of variables, One –dimensional wave equation, the D'Alembert's solution, one-dimensional heat equation, two-dimensional heat flow in steady state (solution of two-dimensional Laplace equation in Cartesian coordinates only)

UNIT-III Difference Equations And Z-Transforms

Formation of a difference equation, Rules for finding complimentary function and particular integral for linear difference equations.

Definition of Z- transform, some standard Z- transforms, properties, transform of a function multiplied by n, initial value theorem and final value theorem(without proof), evaluation of inverse Z- transforms, convolution theorem (without proof), solution of linear difference equations by the use of Z- transforms.

UNIT-IV Probability Distributions

Binomial distribution, Poisson distribution, Normal distribution: Definition (pmf/pdf), notation, mean, variance, moment generating function, probability generating function and fitting of a distribution.

UNIT-V Sampling Theory

Sampling theory: Sampling distribution, standard error, testing of Hypothesis, level of significance, confidence limits, simple sampling of attributes, sampling of variables, estimation of mean and variance.

Large samples: testing of hypothesis for sample proportion, two proportions, single mean and two means.

Small samples: Degrees of freedom, Students' t- distribution, t-test for single mean, two means; Chi-squared distribution-testing the goodness of a fit.

Text Book:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National PublishingCompany.

ELECTRONIC DEVICES AND CIRCUITS
(Common to ECE & EEE)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives:

1. To give exposure to the students about intrinsic and extrinsic semiconductors, semiconductor diodes, special purpose diodes like Zener diode, Photo diode, LED, Schottky barrier diode, PIN diode, varactor diode and tunnel diode etc.
2. To give exposure to the students about rectifier circuits using diodes.
3. To give exposure to the students on basics of BJT, JFET and MOSFET and biasing of BJT and FETs.
4. To give exposure to the students on the analysis of transistor at low and high frequencies.

Course outcomes:

After completion of the course the students will be able to

1. Understand the physical structure, principles of operation, electrical characteristics and circuit models of diodes, BJT's and FET's.
2. Use the concepts of semiconductor physics and electronic devices to design and fabricate simple electronic circuits.
3. Use this knowledge to analyze and design amplifier circuits and oscillator circuits to be used in various applications.
4. Extend the understanding of how electronic circuits and their functions fit into larger electronic systems.

SYLLABUS

UNIT-I: Transport Phenomena in Semi Conductors

Mobility and conductivity, intrinsic and extrinsic semiconductors, mass action law, charge densities in a semiconductors, Hall Effect, generation and recombination of charges, drift and diffusion currents, the continuity equation, injected minority carrier charge, potential variation in graded semiconductors.

UNIT- II: PN junction diode and Diode Rectifiers

Open circuited PN junction , PN junction as a rectifier, current components in a PN diode, V-I characteristics and its temperature dependence, transition capacitance, charge control description of a diode, diffusion capacitance, junction diode switching times, Zener diode, Tunnel Diode, Photo diode,Varactor diode, LED,Half wave, Full wave and Bridge Rectifiers with and without filters, Ripple factor and regulation characteristics

UNIT – III: Bipolar junction transistors

Introduction to BJT, operation of a transistor and transistor biasing for different operating conditions, transistor current components, transistor amplification factors: α, β, γ relation between α and β, γ early effect or base-width modulation, common base configuration and its input and output characteristics, common emitter configuration and its input and output characteristics, common collector configuration and its input and output characteristics, Comparison of CE, CB and CC Configurations, Break- down in transistors, Photo Transistor.

Transistor Biasing Circuits: The operating point, Bias stability, different types of biasing techniques, stabilization against variation in I_{co} , V_{BE} , & β . Bias compensation, thermal runaway, thermal stability.

UNIT – IV: Field Effect transistors

JFET and its characteristics, pinch off voltage, FET small signal model, MOSFET and its characteristics, Biasing of FETs.

UNIT – V: Transistors at low and High frequencies

Transistor hybrid model, H-parameters, Analysis of transistor amplifier circuits using h-parameters, comparison of transistor amplifier configurations, analysis of single stage amplifier, effects of bypass and coupling capacitors, frequency response of CE amplifier, Emitter follower, High frequency model of transistor.

Text books:

1. Integrated Electronics: Analog and Digital Circuits and Systems: Jacob Millman, C Halkias, Chetan D Parikh. McGraw – Hill.
2. Electronic Devices and Circuits: N Salivahanan and Suresh Kumar, Third edition, TMH.

Reference Books:

1. Electronic Devices and Circuits Theory, Boylsted, 10th Edition, Pearson
2. Electronic Principles : Albert Paul Malvino, McGraw-Hill.

NETWORK ANALYSIS AND SYNTHESIS

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

1. To learn various theorems and techniques in electric circuit analysis and to know their significance and applications.
2. To learn the concept of transient analysis of the electric circuit.
3. To learn the concept of Laplace transform and its application to circuit analysis.
4. To learn the concept of two-port network analysis and find models using different parameter sets.
5. To learn the concept of Network functions and the synthesis procedure for RC, LC, RL & RLC networks.

Course Outcomes:

1. Students will learn the theorems for Analyzing complex networks.
2. Students will outline the significance of energy storing elements (Inductance & Capacitance) in circuits and study transient behavior of responses.
3. Students will learn to apply Laplace transform technique for circuit analysis and know its advantages.
4. Students will learn to apply two-port network analysis for devices like amplifiers, transmission lines.
5. Students will learn to apply the concept of positive real functions and the synthesis procedure for RC, LC, RL & RLC networks.

SYLLABUS**UNIT-I: Network theorems**

Linearity and Superposition theorem, Thevenin's theorem, Norton's theorem, Maximum Power Transfer theorem, Reciprocity theorem, Millman's theorem, Tellegen's theorem and Compensation theorem.

UNIT-II DC Transients

Inductor, Capacitor, Source free RL, RC and RLC Response, Evaluation of Initial conditions, application of Unit-step Function to RL, RC and RLC Circuits (Differential equations), Concepts of Natural, Forced and Complete Response.

UNIT-III Laplace Transform Techniques

Transforms of Typical Signals, Response of Simple Circuits to Unit – Step, Ramp and Impulse Functions, Initial and Final Value Theorem, Convolution Integral, Time Shift and Periodic Functions, Transfer Function

UNIT-IV Two Port Networks

Two port network parameters – Z, Y, ABCD and Hybrid parameters and Interrelationship between different parameters.

UNIT-V Network Functions & Synthesis

Network Functions ,Concept of Poles and Zeroes, Restriction of Poles and Zeroes for Driving point and transfer function, Hurwitz Polynomial.

Positive real function - basic synthesis procedure - Foster and Cauer forms of LC, RC and RL networks.

Text Books:

1. Engineering Circuit Analysis, William H.Hayt Jr. and Jack E. Kemmerley, 5th Edition, McGraw Hill International Edition.
2. Introduction to Modern Network Synthesis ,Van Valkenburg; John Wiley

Reference Books:

1. Network analysis,M.E. Van Valkunberg, 3rd Ed., phi/Eee pub.
2. Network Analysis &Synthesis ,F.F.Kuo; John Wiley & Sons Inc.
3. Circuit Theory Analysis and Synthesis., Edition 2014 Abhijit Chakrabarthi,DhanpatRai&Co.
4. Network Analysis, 3rd Ed, A SudhakarShyammohan.SPalli Tata McGraw Hill Education Pvt Ltd.

ELECTRO MAGNETIC FIELD THEORY

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

1. All the electric equipment is developed by using the magnetic material, conductors and insulators. It is very much essential to know the behaviour of these materials in the presence of electric and magnetic fields.
2. The main objective of this course is to provide the basic concepts about the effects of electric and magnetic fields on conductors, magnetic materials, and insulators under various operating conditions.

Course Outcomes:

Students are able to

1. Find the electrostatic and magneto static fields for different configurations.
2. Apply various principles and laws to estimate the effect of electric and magnetic fields.
3. Distinguish between the effects of electrostatic and magneto static fields.
4. Apply Maxwell's equations for static and time varying fields.
5. Analyze the EM wave in different domains and compute average power density

SYLLABUS**UNIT I****Coordinate systems:**

Rectangular, cylindrical and spherical coordinate systems.

Electrostatics:

Coulomb's law and superposition principle, different types of charge configurations, electric flux, electric field intensity and electric flux density, electric field intensity and electric flux density due to different charge configurations, Gauss's law in integral form and point form in terms of D , applications of Gauss' law, Divergence theorem.

UNIT II

Electric potential, calculation of electric potential for given charge configuration, electrostatic energy, Electrostatic boundary conditions, basic properties of conductors in electrostatic fields, capacitance, Poisson's and Laplace's equations, solutions of Laplace's equations, uniqueness theorems, methods of images, electric dipoles, polarization of dielectrics, bound charges.

UNIT III**Magneto statics:**

Biot-savart's law, determination of magnetic field intensity and magnetic flux density due to various steady current configurations, continuity equation, curl of $\vec{H}$, Ampere's circuital law in integral and differential form, applications of Ampere's law, Stokes theorem.

UNIT IV

The scalar and vector magnetic potential and calculation of magnetic field through the vector magnetic potential for given steady current configurations, magnetostatic boundary conditions. The magnetic dipole, magnetization, properties of magnetic materials, torques and forces on magnetic dipoles, bound current, Faraday's laws, Lenz's law, inductance and energy in magnetic fields.

UNIT V

Time varying fields and Maxwell's equations:

Lorentz force equation, Maxwell's equations, modification of ampere's circuital law for time varying fields – displacement current and current density, the uniform plane wave, plane wave propagation, phase velocity and wavelength, intrinsic impedance, attenuation, phase and propagation constants, skin depth, the poynting vector, poynting theorem and power considerations.

Textbooks:

1. Introduction to electro dynamics by D.J. Griffiths, PHI Learning; 3rd Edition (2012).
2. Engineering electromagnetics by William H. Hayt , John A. Buck McGraw-Hill Publishing Co. (2001).

Reference books:

1. Principles of Electromagnetics by [Mathew N.O. Sadiku](#), Oxford; Fourth edition.(2009).

ELECTRICAL MEASUREMENTS & INSTRUMENTS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives

1. The students learn about measuring instruments to measure electrical quantities like current, voltage etc.
2. This course familiarizes with Wattmeter's, Energy meters, power factor meters, frequency meters etc.
3. The students learn to measure resistance, inductance, capacitance, etc. using bridges.
4. The course familiarizes ballistic galvanometer, flux meter, testing of ring and bar specimens for magnetic measurements.
5. Students learn about AC & DC potentiometers, phase & amplitude measurements, use of a CRO, instrument transformers.

Course Outcomes

Upon successful completion of this course, the students will be able to:

1. Illustrate the characteristics of measuring instruments (K3)
2. Discriminate measuring instruments based on their principle & operation (K4)
3. Calculate power and energy in 1 ϕ , 3 ϕ & polyphase circuits (K3)
4. Measure electrical parameters using a bridge (K3)
5. Find magnetic measurements using Ballistic Galvanometers and Flux meters. (K4)
6. Apply potentiometers & instrument transformers to measure electrical elements, calibration of the meters.

SYLLABUS**UNIT-I:Philosophy of measurement**

Methods of measurement, measurement system, classification of instrument system, characteristics of instruments & measurement system, errors in measurement & its analysis, standards.

UNIT-II:Analog measurement of electrical quantities

Moving coil, moving iron, Electrodynamometer type, electrostatic and induction type instruments, electrodynamic wattmeter, three phase wattmeter, power in three phase system, errors & remedies in wattmeter and energy meter. Extension of instrument range, introduction to measurement of frequency and power factor.

UNIT-III:Measurement of parameters

Different methods of measuring low, medium and high resistances, measurement of inductance & capacitance with the help of AC bridges. DC potentiometers and its applications. AC potentiometer - types & applications.

UNIT-IV: Magnetic measurement

Ballistic galvanometer, flux meter, determination of B-H curve and hysteresis loop, measurement of iron losses, current transformers and potential transformers, application of CRO in measurement of B-H curve.

UNIT-V: Digital measurement of electrical quantities

Digital Instruments, Concept of digital measurement, Analog to digital & Digital to analog conversion, advantages of digital Instruments, digital display units, Resolution in digital meters, sensitivity & Accuracy of digital meters.

Text Books:

1. E.W. Golding & F.C. Widdis, "Electrical Measurement & Measuring Instrument", Reem Publications Pvt. Ltd.; Third edition (2011).
2. A.K. Sawhney, "Electrical & Electronic Measurement & Instrument", Dhanpat Rai & Co. (P) Limited; 2014 edition (2015)
3. W.D. Cooper, "Electronic Instrument & Measurement Technique", Prentice Hall International.

Reference Books:

1. Forest K. Harries, "Electrical Measurement", Willey Eastern Pvt. Ltd. India.
2. M.B. Stout, "Basic Electrical Measurement", Prentice Hall of India.
3. Rajendra Prashad, "Electrical Measurement & Measuring Instrument", Khanna Publisher.
4. J.B. Gupta, "Electrical Measurements and Measuring Instruments", S.K. Kataria & Sons, 2012 Edition.

ENGINEERING ECONOMICS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objective:

To create awareness on application of economic & accounting concepts in the organization for engineering students. In this regard, they will go through demand analysis, cost concepts, pricing practices, macro- economic concepts & fundamentals of accounting.

Course Outcomes:

1. Provide detailed insight about origin & definitions of economics & enlighten the students about demand analysis.
2. Illustration about applications of cost Concepts & analysis of breakeven point.
3. Understand about various types of Market Structure and Pricing practices implemented by the organization.
4. Infuse knowledge about different Economic systems & Business cycles.
5. Enlighten the students regarding the aspects of Depreciation & Financial Accounting.

SYLLABUS**UNIT-I**

Introduction to Economics: Wealth, Welfare and Scarce Definitions of Economics; Micro & Macro Economics.

Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of demand – Meaning, types, Significance of Elasticity of Demand, Measurement of price Elasticity of Demand. Need for Demand forecasting, forecasting techniques.

UNIT-II

Cost Analysis: Classification of cost, Elements of cost, Methods of costing (Job costing, Process costing & Unit costing).

Break-Even Analysis(BEA): Determination of Break-Even Point, Assumptions and Applications.

UNIT-III

Market Structures: Features and price determination under Perfect competition, Monopoly, Monopolistic competition and Oligopoly.

Pricing practices: Price - meaning, methods of pricing.

UNIT-IV

Economic Systems: Features and Evaluation of Capitalism, Socialism and Mixed Economy.

Business cycles: Meaning, Phases , Causes & theories of Business Cycle.

UNIT-V

Depreciation and Financial Accounting: Depreciation-causes and methods (straight line method, diminishing balance method).

Final Accounts: Preparation of Trading Account, Profit & Loss Account and Balance sheet.

Text Books:

1. Managerial Economics & Financial Analysis-by Dr.A.R.AryaSri,TMH 2011.
2. Engineering Economics-by Tarachand,Nem Chand &Bros.Roorke.

Reference Books:

1. Modern Economics - by K. K. Dewett,S.Chand&Co,New Delhi.
2. Principles of Economics-Vrinda Publications(P)Ltd.New Delhi.

NETWORKS & MEASUREMENTS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To learn to make simple electric circuits by using different sources, loads and components and verify basic laws.
2. To experimentally verify various theorems of circuit analysis.
3. To learn to find circuit models for two-terminal devices and two-port networks.
4. To explore the sinusoidal steady state behavior and resonance phenomenon in electric circuits.
5. To measure different electrical parameters by using different meters.

Course Outcomes:

1. Students will gain the skill to make and experiment with practical electric circuits.
2. Students will be able to measure voltage, current, power in practical electric circuits.
3. Students will know the significance of various theorems and their applications.
4. Students will be able to assess the behavior of electric circuits.
5. Students will be able to calibrate single phase energy meter, voltmeter & wattmeter
6. Students will be able to measure resistance, inductance & capacitance.

LIST OF EXPERIMENTS

- 1.Verification of Ohms Law and resistance of a filament Lamp
- 2.Verification of superposition theorem
- 3.Verification of Thevenin's theorem
- 4.Verification of Norton's theorem
5. Verification of maximum power transfer theorem
- 6.Series resonance
7. Calculation two port network parameters
- 8.Calibration of wattmeter
- 9.Calibration of energy meter
- 10.Three voltmeter method
- 11.Measurement of 3 phase power using two wattmeter method
- 12.Parameters of choke coil.
- 13.Measurement of three phase power by using 2 C.T's and Single Wattmeter
- 14.Crompton's DC potentiometer
- 15.Kelvin's double bridge
- 16.Schering bridge

Reference Books:

1. A.K. Sawhney, "Electrical & Electronic Measurement & Instrument", Dhanpat Rai & Co. (P) Limited; 2014 edition (2015).
2. Engineering circuit analysis by W.H. Hayt Jr & J.E. Kemmerly, McGraw Hill Education; Eighth edition (4 August 2013).

ELECTRONICS DEVICES AND CIRCUITS LAB
(Common to ECE & EEE)

Lab : 3 Periods

Exam : 3Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course objectives:

1. To familiarize the students with various passive and active components like resistors, capacitors, inductors, semiconductor diodes, Zener diodes, LEDs, BJTs, JFETs and UJT.
2. To familiarize the students with operation of CROs, function generators and bread boards.
3. To observe and analyze the characteristics of devices like diodes, BJTs & FETs.
4. To analyze the behavior of BJT and JFET amplifiers.

Course outcomes:

At the end of the semester students should be able to

1. Design and fabricate simple circuits like diode rectifiers with filters for providing dc voltages in electronic circuits.
2. Design and fabricate amplifiers with required gain for use in various communication applications.
3. Design and fabricate simple electronic circuits for everyday applications like traffic control lights using relays, automatic counters using LDRs and Burglar alarms.

ELECTRONIC WORKSHOP PRACTICE

1. Identification, Specifications and testing Of R,L,C components, colour codes, potentiometers, coils and bread boards
2. Identification, Specifications and testing of devices like diodes, BJTs, JFETs, SCR and UJT.
3. Soldering of Simple Circuits using Active & Passive Components.
4. Study and operation of Transformers, Ammeters (Analog & Digital), Voltmeters (Analog & Digital), Analog and Digital Multimeters and Function Generators, Regulated Power Supply, Decade Resistance, Inductance & Capacitance Boxes And CRO.

LIST OF HARDWARE EXPERIMENTS:

1. V-I Characteristics Of Semiconductor Diode (Ge & Si), LED and Zener Diode
2. Half Wave And Full Wave Rectifier With And Without Filter
3. Characteristics Of BJT In CE Configuration
4. JFET Characteristics
5. Transistor Biasing Circuits And Transistor As Switch
6. CE Amplifier
7. JFET Common Source Amplifier

LIST OF SIMULATION EXPERIMENTS

1. Simulation of V-I Characteristics Of Semiconductor Diode, LED and Zener Diode
2. Simulation of Regulation Characteristics Of ZENER Diode
3. Simulation of CC Amplifier
4. Simulation of JFET Characteristics
5. Simulation of BJT Characteristics In CB Configuration
6. Simulation of JFET Amplifier
7. Simulation of UJT Characteristics

NOTE: (Minimum of Twelve Experiments Should Be Conducted)

Reference Books:

1. Integrated Electronics: Analog and Digital Circuits and Systems: Jacob Millman, C Halkias, Chetan D Parikh. McGraw – Hill.

**PROGRAMMING SKILLS-I
(PYTHON)**
(Common to ECE & EEE)

Lab	: 2 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Credits	: 1

Course Objectives:

1. To know basic principles of Object Oriented Programming in the context of python programming language.
2. To study different data types of python, to design a programs.
3. To study and apply different types of concepts like Inheritance, Exception Handling, Turtle and design programs using these concepts.
4. To study different built in modules like sys, os , math & SQLite
5. To study server side scripting in python like CGI scripts

Course Outcomes:

1. Ability to apply object oriented concepts in programming.
2. Ability to define, understand and differentiate different types of data types and apply them.
3. Ability to recognize various concepts of python and develops the programs using them and also develop web based application.

SYLLABUS

UNIT-I:

Overview, Environment Set Up, Basic Syntax, Identifiers, Reserved Words, Lines and Indentation, Multi-Line Statements, Quotation, Comments, Multiple Statements on a Single Line Variable Types, Standard Data Types, Numbers (math, random, fraction) , Strings, Lists, Tuples , Dictionaries

UNIT-II:

Operators, Arithmetic Operators, Comparison (Relational) Operators, Assignment Operators, Logical Operators, Bitwise Operators, Membership Operators, Identity Operators, Decision Making :if, if-else, nested if , Loops: for, while, nested loops

UNIT-III:

Functions, Function Arguments: Required arguments, Keyword arguments, Default arguments, Variable-length arguments, The Anonymous Functions: lambda, Scope of Variables, Modules, sys, os , Date & Time

UNIT-IV:

Files & its operations, Exceptions, Standard Exceptions, Assertions, The try-finally Clause, Raising an Exception, User-Defined Exceptions, Classes and objects , OOPS, Data member , Function overloading, Instance variable, Inheritance, Instance, Instantiation, Operator overloading

UNIT-V:

HTML,CSS Basics, Data Base(SQLite), Database Connection, CRUD Application , CGI Architecture, Web Server Support and Configuration, GET and POST Methods, CGI Scripts.

UNIT-VI:

Project

References:

1. Dive Into Python 3
2. Think Python
3. Halterman python book
4. Dr. Andrew N. Harrington Computer Science Department, Loyola University Chicago©
Released under the Creative commons Attribution-Noncommercial-Share Alike 3.0
United States License <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>
5. <https://www.python.org/>
6. <http://www.tutorialspoint.com/python3/>
7. <https://www.w3schools.com/>

ENGLISH PROFICIENCY-I
(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

The Students will be able to

1. Communicate their ideas and views effectively.
2. Practice language skills and improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self attributes which require improvement and motivation.
5. Build confidence and overcome their inhibitions, stage freight, nervousness, etc.
6. Improve their innovative practices in speaking.

Course Outcomes:

The Students will

1. Improve speaking skills.
2. Enhance their listening capabilities.
3. Learn and practice the skills of composition writing.
4. Enhance their reading and understanding of different texts.
5. Improve their inter-personal communication skills.
6. Be confident in presentation skills.

SYLLABUS

UNIT-1:LISTENING

Selected Motivational Speeches
Selected Moral Stories

UNIT-2:SPEAKING

Book Review
Skit Presentation
PowerPoint Presentations
Describing event/place/thing
Extempore
Group Discussion
Picture Perception and Describing Test

UNIT-3:READING

Speeded Reading
Reading Comprehension

UNIT-4:WRITING

Paragraph Writing
Literary Appreciation – Understanding the Language of Literature

UNIT-5:PROJECT

Ad Making

References:

1. Classic Short Stories-A Reader's Digest Selection
2. English for Colleges by Brendan J. Carroll, Macmillan Publications
3. The World's Great Speeches- edited by B.S. Sekhar, Jeet Publications
4. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
5. English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
6. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
7. The Oxford Guide to Writing and Speaking by John Seely OUP

PROFESSIONAL ETHICS & HUMAN VALUES

(Common to CIVIL,EEE & MECH)

Lecture	: 2 Periods.	Int.Marks	: --
Exam	: ---	Credits	: --

Course Objectives:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

Course outcomes:

By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS

UNIT – I

Ethics and Human Values: Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

UNIT – II

Engineering Ethics: Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics

Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

UNIT – III

Engineering as Social Experimentation:

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders Role of engineers in promoting ethical climate, balanced outlook on law.

UNIT – IV

Safety Social Responsibility and Rights:

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

UNIT – V

Global Issues:

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

Text Books:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

References:

1. Charles D, Fleddermann, “Engineering Ethics”, Pearson / PHI, New Jersey 2004 (Indian Reprint).

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

II/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)

ELECTRICAL AND ELECTRONICS ENGINEERING
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/Week	Internal Marks	External Marks	Total Marks
B17EE 2201	Electrical Machines-I	3	3	1	--	4	30	70	100
B17EE 2202	Signals & Systems	3	3	1	--	4	30	70	100
B17EC 2205	Electronic Circuit Analysis	3	3	1	--	4	30	70	100
B17ME 2206	Prime movers & Pumps	3	3	1	--	4	30	70	100
B17EC 2206	Linear Integrated and Pulse Circuits	3	3	1	--	4	30	70	100
B17BS 2201	Management Science	3	3	1	--	4	30	70	100
B17ME 2210	Thermal Prime Movers Lab	2	--	--	3	3	50	50	100
B17EC 2209	Linear Integrated Circuits & Pulse Digital Circuits Lab with Simulation	2	--	--	3	3	50	50	100
B17BS 2205	Programming Skills-II	1	--	--	2	2	50	---	50
B17BS 2206	English Proficiency-II	--	1	1	--	2	--	--	--
Total		23	19	7	8	34	330	520	850

ELECTRICAL MACHINES-I

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

The course will enable the students to understand

1. Electro-mechanical energy conversions in D.C. machines and energy transfer in transformers
2. Principle of operation of DC machines and transformers
3. Speed control methods of DC motors and parallel operation, testing of DC machines and transformers.
4. Different types of three phase transformer connections.

Course Outcomes:

Students are able to

1. Identify the concepts of electro mechanical energy conversion. [K2]
2. Describe the concepts of construction, operating principle, different types of DC machines and transformers, effects on DC machine and parallel operation of DC generators. [K2]
3. Interpret the characteristics of DC machines. [K3]
4. Discriminate different types of speed control methods of DC motors. [K4]
5. Examine the performance of DC machines and transformers by different testing methods. [K4]
6. Discriminate different types of transformer connections [K4]

SYLLABUS**UNIT-I: Electromechanical energy conversion:**

Basic principles of energy, force and torque in singly and multiply excited systems.

UNIT-II: Transformers:

Principle, construction and operation of single phase transformers, phasor diagram, equivalent circuit, voltage regulation, losses and efficiency. Testing- open & short circuit tests, Sumpner's test.

Autotransformers- construction, principle, applications and comparison with two winding transformer.

UNIT-III: Three phase transformer:

Construction, various types of connection and their comparative features. Parallel operation of single phase and three phase transformers. Three phase transformer connections. Scott connection, tap changing transformers- no load and on load tap changing of transformers. Cooling methods of transformers.

UNIT-IV: D.C. Machines-

Working principle, construction and methods of excitation. D.C generators emf equation, armature reaction, commutation. Compensating winding, characteristics of various types of generators, applications. D.C. motors- torque equation, D.C. shunt, series and compound motors – characteristics& applications.

UNIT-V: Starting &Speed control-

Starting methods and speed control of D.C. shunt and series motors testing of D.C motors - direct and regenerative methods to test D.C. machines. Swinburne's test, field's test and separation of losses.

Text books

1. Kothari.D.P and Nagrath.I.J., “Electrical machines”, McGraw Hill Education; 4 edition (2010).
2. Bimbhra.P.S, Electrical Machinery, Khanna Publishers, 2011.
3. [Irving L. Kosow](#), “Electrical Machines & Transformers”, Prentice Hall; 2nd Revised edition 1990.

Reference Books

1. Clayton. A.E,,,Performance and Design of direct current machines“ CBS; 1ST edition (2004).
2. Mg Say, theory, ”Performance & Design of A.C Machines”, CBS publishers.
3. Fitzgerald, A.E., Charles Kingselyjr. Stephen D.Umans, “electric machinery”McGraw-Hill; 6th edition (2005).
4. Hill Stephen, Chapman.j, “Electric Machinery Fundamentals”, McGraw-Hill Higher Education; 4 edition (2004).

SIGNALS & SYSTEMS

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives

1. Coverage of continuous and discrete-time signals and systems, their properties and representations and methods those are necessary for the analysis of continuous and discrete-time signals and systems.
2. Knowledge of time-domain representation and analysis concepts as they relate to difference equations, impulse response and convolution, etc.
3. Knowledge of frequency-domain representation and analysis concepts using Fourier Analysis tools, Z-transform
4. Concepts of the sampling process.

Course Outcomes:

Upon the completion of the course, students will be able to:

1. Characterize and analyze the properties of continuous and discrete time signals and systems. [K2]
2. Apply the convolution for continuous time signals and discrete time signals. [K3]
3. Evaluate the Fourier Series of periodic signals. [K1]
4. Determine the Fourier Transform and Z-Transform of different type's of signals and make use of their Properties. [K1]
5. Convert a continuous time signal to the discrete time domain and reconstruct using the sampling theorem. [K2]

SYLLABUS**UNIT I Classification of Signals & Systems:**

Basic continuous time signals, basic discrete time signals transformations of independent variables, classification of systems, properties of linear time – invariant systems.

UNIT II Linear Time – Invariant (LTI) Systems:

Representation of signals in terms of impulses for discrete time and continuous time signals, convolution sum and convolution integral. systems described by differential and difference equations. Block diagram representation of LTI systems described by differential and difference equations, singularity functions.

Analogy between vectors and signals, orthogonal vector and signal spaces. Approximation of a function by a set of mutually orthogonal functions.

UNIT III Fourier analysis:

The response of continuous time LTI systems to complex exponentials – the continuous time and discrete time exponential fourier series, convergence of fourier series.

UNIT IV Fourier Transform:

Fourier transform of continuous time and discrete time aperiodic signals and periodic signals. properties of continuous time and discrete time fourier transforms. Frequency response characterized by linear constant coefficient differential and difference equations. first order and second order systems.

UNIT V: Z –transform:

Z–transform of discrete time sequence, region of convergence. relation between Z and fourier transform, properties of z-transforms. inverse z-transform, determination of transfer function and impulse response of an LTI system, poles and zeros and system stability.

Sampling Theorem :

The effect of under-sampling, methods of reconstruction of a signal from samples, discrete time processing of continuous time signals.

Text Books:

1. Signals and Systems, Alan V. Oppenheim, Alan S. Willsky and Ian T. Young, Prentice-Hall; New edition (1984).

Reference Books:

1. Communication Systems, B. P. Lathi., BS PUBLICATION (2001).
2. Signals and Systems, B. P. Lathi., Oxford University Press; 2nd edition (2004).

ELECTRONIC CIRCUIT ANALYSIS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The aim of this course is to

1. Understand the concept of multistage amplifiers and analyze them.
2. Learn the classification of feedback amplifiers and analyze them.
3. Compare voltage, power and tuned voltage amplifiers and analyze them.
4. Understand the principle of oscillator and analyze different types of sinusoidal oscillators.
5. Understand the concept and analyze applications of op-amp.

Course Outcomes:

After the completion of the course, students will be able to:

1. Know the equivalent circuit of multistage amplifier and its analysis. [K3]
2. Identify the different feedback topologies and analyze them. [K1]
3. Explain the principle of oscillator and design different types of sinusoidal oscillators. [K3]
4. Define the difference between voltage and power amplifiers and design different classes and know that Tuned amplifiers amplify a narrow band of frequencies and will also be able to analyze them.[K1,K2, K3]
5. Identify that Op-amp not only amplifies but also performs different operations and analyze some of its applications.[K1,K2]

SYLLABUS**UNIT – I: Multistage Amplifiers**

Transistor at high frequencies, CE short circuit current gain and concept of Gain Bandwidth product. BJT and FET RC coupled amplifiers at low and high frequencies. Frequency response and calculation of Band Width of Multistage Amplifiers.

UNIT – II: Feed Back Amplifiers

Concept of Feed Back Amplifiers - Effect of Negative Feedback on the amplifier characteristics. Four feedback topologies, Method of analysis of Voltage Series, Current Series, Voltage Shunt and Current Shunt feedback Amplifiers.

UNIT – III: Sinusoidal Oscillators

Condition for oscillations and types of Oscillators – RC Oscillators: RC Phase Shift and Wien bridge Oscillators. LC Oscillators: Hartley, Colpitts, Clapp, Tuned Collector and Crystal Oscillators.

UNIT – IV: Power and Tuned Voltage Amplifiers

Classification of Power Amplifiers. Series fed, Transformer coupled class-A and class-B power amplifiers. Push Pull Class-A, Class-B and Class-AB Power Amplifiers. Cross-over Distortion in Pure Class-B Power Amplifier and Class-AB Power Amplifier- Trickle Bias, Derating Factor and Heat Sinks – Complementary Push Pull Amplifier. Analysis of Single tuned, Double tuned and Stagger Tuned Amplifiers with gain and Bandwidth Calculations.

UNIT – V: Operational Amplifiers

Concept of Differential Amplifier. Differential Amplifier supplied with a constant current source. Calculation of common mode rejection ratio. Block diagram and Ideal characteristics of an Op-Amp. Applications of Op-Amp: Inverting and Non-Inverting amplifiers, Integrator, Differentiator, Summing, Subtracting and Logarithmic Amplifiers. Definition and Measurement of OP-Amp Parameters.

Text Books:

1. Integrated Electronics- Millman and Halkias.
2. Electronic Devices and Circuits – Mottershead.
3. Op-amps and Linear Integrated Circuits – Gayakwad.

Reference Book:

1. Electronic Devices and Circuits by Salivahanan. Tata McGraw-Hill pub.

PRIME MOVERS & PUMPS

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

The objectives of the course are:

1. To make the students understand the various types of prime movers which can be connected to generators for power production
2. To make the student learn about the constructional features, operational details of various types of internal combustion engines through the details of several engine systems and the basic air standard cycles, that govern the engines.
3. To train the student in the aspects of steam formation and its utilities through the standard steam data tables and chart
4. To make the student correlate between the air standard cycles and the actual cycles that govern the steam turbines
5. To teach the student about the fundamental of fluid dynamic equations and its applications fluid jets
6. To impart the knowledge of various types of pumps.

Course Outcomes:

After the completion of the course, students are able to

1. Understand the concepts of hydrodynamic force of jets on stationary and moving flat inclined and curved vanes.
2. Apply the concepts of momentum equation for finding the forces acting on the vanes of the turbines.
3. Calculate the performance of different types of internal combustion engines.
4. Apply the otto, Diesel cycles for finding the performance of S.I and C.I engines. Understand the working principle of steam turbines and gas turbines.
5. To impart the knowledge of various types of pumps, their constructional features, working and performance.
6. To impart the knowledge of various types of turbines and the performance characteristics of hydraulic turbines

SYLLABUS**UNIT-I I.C Engines:**

Classification, working principles – valve and port timing diagrams – air standard cycles :otto, diesel -P-V and T-S diagram ,thermal efficiency– Engine systems line fuel injection, carburetion, ignition, cooling. Engine performance evaluation.

UNIT-II Properties of Steam and use of Steam Tables:

T-S and H-S Diagrams. Analysis of Various Thermodynamic Processes under gone by Steam.
Vapor Power Cycles: Carnot Cycle-Rankine Cycle- Thermodynamic Variables Effecting Efficiency and output of Rankine Cycle-. Analysis of simple Rankine Cycle and Re-heat cycle.

UNIT- III Gas Turbines:

Simple gas turbine plant-ideal cycle, closed cycle -open cycle-. Efficiency, Work ratio and optimum pressure ratio for simple gas turbine cycle. Actual cycle, analysis of simple cycles & cycles with inter cooling, reheating

Steam Turbines: Classification of Steam Turbines Impulse Turbine and Reaction Turbine- Compounding in Turbines- Velocity Diagrams for simple Impulse and Reaction Turbines- Work done & efficiency

Part-B: Hydro prime movers

UNIT-IV Impact Of Jets And Pumps:

Impulse momentum equation, Impact of Jet on stationary and moving vanes (flat and curved).

PUMPS: Types of pumps, Centrifugal pumps: Main components, Working principle, Multi stage pumps, Performance and characteristic curve

UNIT-V: Hydraulic Turbines:

Classification of turbines; Working principle, Efficiency calculation and Design principles for Pelton Wheel, Francis and for Kaplan turbines; Governing of turbines; Performance and characteristic curves.

Text Books:

1. Thermal Engineering by Rajput, Lakshmi publications
2. Thermal engineering by M.L.Mathur and F.S.Mehta, Jain Brothers.
3. "Hydraulics & Fluid Mechanics", P.N. Modi and S.M. Seth, TEXT BOOKS House, Delhi
4. "Fluid Mechanics & Hydraulic Machinery" A.K.Jain, , Khanna Publishers, Delhi.

Reference Books:

1. Fluid Mechanics & Hydraulic Machines- by R.K.Bansal, Laxmi Publications.
2. Thermodynamics & Heat Engines – by B.Yadav, Central Book Depot, Allahabad.
3. I C Engines – by V Ganeshan, Tata McGraw-Hill Companies.
4. "Fluid Mechanics & Its Applications", Vijay Gupta, Santhosh.k.Gupta
5. "Fluid Mechanics & Fluid power Engineering, Dr D.S.Kumar

LINEAR INTEGRATED AND PULSE CIRCUITS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students should learn

1. Different applications Op-amps
2. Design of active filters and oscillators
3. Study of 555 Timers and IC regulators
4. Applications of integrator, differentiator, clippers and clamper circuits.
5. Design of various multivibrators for various applications.

Course Outcomes:

Students will be able to

1. Understand the applications of Op-amps
2. Design different active filters and oscillators
3. Understand the applications of 555 Timers and IC regulators
4. Understand the applications of integrator, differentiator, clippers and clamper circuits.
5. Design different multi vibrators for various applications.

SYLLABUS**UNIT-I: Applications of Operational Amplifiers:**

Basics of Op-Amp, Instrumentation Amplifiers, Voltage to Current and Current to Voltage Converters. Op-amp As a Comparators, Schmitt trigger, Wave form Generators, Sample and Hold Circuits, Rectifiers.

UNIT-II: Active Filters and Oscillators:

Butterworth type LPF, HPF first and second order filters, Switched Capacitance Filters. Op-Amp Phase Shift, Wein-bridge and Quadrature Oscillator, Analog Multiplexers.

UNIT-III: Special ICs:

555 Timers Introduction, Block diagram, 555 timer as an astable and Monostable Multivibrator, Three Terminal IC Regulators, Voltage to Frequency and Frequency to Voltage Converters.

UNIT-IV: Wave Shaping:

High pass and Low pass RC circuits, Response of High pass and Low pass RC circuits to step, square inputs. High pass RC circuit as a differentiator, Low pass RC circuit as an integrator. Diode clippers, Clipping at two independent levels, Clamping Operation, Clamping Circuits using Diode with Different Inputs, Clamping Circuit Theorem, Practical Clamping circuits.

UNIT-V: Multivibrators:

Transistor as a switch, Switching times of a transistor, Design and Analysis of Fixed-bias and self-bias transistor binary, Commutating capacitors, Design and analysis of Collector coupled Monostable Multivibrator, Expression for the gate width and its waveforms. Design and analysis of Collector coupled Astable Multivibrator, Expression for the Time period and its waveforms, The Astable Multivibrator as a voltage to frequency converter.

Text Books:

1. Microelectronics- Jacob Millman.
2. Op-Amps and Linear ICs- RamakanthGayakwad, PHI, 1987.
3. Pulse Digital and Switching Waveforms, J. Millman ,H. Taub, and M.S.PrakashRao McGraw-Hill, Second Edition

Reference Books:

1. Linear Integrated Circuits- D.Roy Chowdhury, New Age International(p) Ltd, 2nd Edition,2003.
2. Pulse and Digital Circuits, A. Anand Kumar, PHI, Second Edition, 2005.

**MANAGEMENT SCIENCE
(Common to ECE & EEE)**

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int.Marks : 30
Ext. Marks : 70
Credits : 3

Course Objective:

To enlighten the technical students with functional management related issues like Principles of Management, Marketing Management, Human Resource Management, Production Management, Financial Management and Strategic Management techniques.

Course Outcomes:

1. Create awareness about the concepts like Evolution of Management thought, functions & principles of management.
2. Provide all round information to the students about matters related to concepts & functions related to Marketing.
3. Acquire in-depth knowledge about the concepts and functions of HRM.
4. Understand about aspects of Production Management and Financial Management.
5. Gain knowledge about Strategy formulation & implementation, SWOT analysis in order to compete with the competition & to gain competency advantage.

SYLLABUS**UNIT-I: Introduction to Management**

Concept, Nature and importance of Management, Functions of management, Evolution of Management thought, Fayol's principles of Management, Theories of Motivation, Decision making process.

UNIT-II: Marketing Management

Concept, Functions of marketing, Marketing Mix, Marketing strategies based on Product life cycle, Channels of distribution.

UNIT-III: Human Resource Management (HRM)

Concepts of HRM, Personal Management and Industrial Relations, Basic functions of HR Manager-Man power planning, Recruitment, Selection, Placement, Training, Development, Compensation and Performance Appraisal.

UNIT-IV: Production Management

Production planning & control (PPC), Objectives, Functions, Stages of PPC, Plant location (Site Selection).

Financial Management

Types of capital- Fixed and Working Capital, Methods of Raising finance. Long-term, Medium-term and Short-term financial sources.

UNIT-V:Strategic Management

Vision, Mission, Goals, objectives, policy, strategy, Elements of corporate planning process, Environmental scanning, SWOT analysis Steps in strategy formulation and implementation of Generic strategy alternatives

Text Books:

1. Dr. Arya Sri – Management Science, TMH 2011

Reference Books:

1. Marketing Management- PHILIP KOTLER.
2. HRM & IR- P.SUBBA RAO
3. Business Policy & Strategic Management- FRANCIS CHERUNILAM
4. Financial Management - I.M.Pandey.

THERMAL PRIME MOVERS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int. Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To understand the principle and functioning of various IC engines.
2. Ability to understand the working of two stroke and four stroke engines.
3. Acquiring the knowledge of operation of a turbines and pumps.
4. The way of determination of flash and fire points of oil samples and their importance is acquired.
5. The procedure for determination of viscosities of oil samples can be understood.

Course Outcomes:

Students will be able to:

1. Explain the working principle of different types of IC Engines and illustrate the valve timing and port diagrams of an IC engines.
3. Determine the viscosities of oil samples, Flash and Fire point values of fuels.
4. Perform the load, Morse, Heat balance and economical speed test on IC Engines.
5. Discuss the working principle of different types of hydraulic turbines
6. Illustrate the working principle of centrifugal and reciprocating pumps

SYLLABUS

1. Drawing of VTD for four-stroke and PTD of two-stroke engines.
2. Determination of flash and fire points
3. Determination of the kinematic and absolute viscosity of the given sample oils.
4. Load test and smoke test on I.C. engines.
5. Morse test on multi-cylinder engine.
6. Heat balance sheet on I.C. engines.
7. Study of multi-cylinder engines and determination of its firing order.
8. Economical speed test on IC engines.
9. Study on impulse and reaction turbines
10. Study on reciprocating and centrifugal pumps

Reference Books:

1. Thermal Engineering, by R. K. Rajput, Lakshmi Publications.
2. Thermal Science and Engineering by D.S. Kumar, S.K. Kataria and Sons.
3. I.C engines by V. Ganesan, Mc Graw Hill Publications.

LINEAR INTEGRATED CIRCUITS & PULSE DIGITAL CIRCUITS LAB WITH SIMULATION

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

This laboratory course enables students to get practical experience in design, assembly and evaluation of Linear integrated circuits & Pulse Circuits. They will use Multisim to test their electronic designs.

Course Outcomes:

Students will be able to:

1. Design and conduct experiments on RC low pass and high pass circuits.
2. Observe operation of UJT Sweep Generator.
3. Design and test different types of Multi vibrators
4. Acquire a basic knowledge on simple applications of operational amplifier.
5. Design, construct Schmitt trigger using operational amplifier.
6. Use Multisim to test their electronic designs.
7. Design and test different types of Multiplexers and counters.

LIST OF EXPERIMENTS

1. Linear Wave Shaping
 - a) Passive RC Differentiator
 - b) Passive RC Integrator
2. Non Linear Wave shaping
 - a) Clipping Circuits
 - b) Clamping Circuits
3. Self bias bistable Multivibrator
4. Schmitt Trigger Using $\mu A 741$
5. UJT Sweep Generator
6. Astable Multivibrator using 555 timer
7. Multiplexer
8. Shift Registers

LIST OF EXPERIMENTS (Simulation)

1. Linear Wave Shaping
 - a) Passive RC Differentiator
 - b) Passive RC Integrator
2. Non Linear Wave shaping
 - a) Clipping Circuits
 - b) Clamping Circuits
3. Self bias bistableMultivibrator
4. Schmitt Trigger Using μA 741
5. UJT Sweep Generator
6. AstableMultivibraotr using 555 timer.
7. Multiplexer
8. Shift Registers

Reference: Lab Manuals

**PROGRAMMING SKILLS-II
(JAVA)
(Common to ECE & EEE)**

Lab : 2 Periods
Exam : 3 Hrs.

Int.Marks : 50
Credits :1

Course Objectives:

1. To know basic principles of Object Oriented Programming in the context of java programming language.
2. To study different types of arrays to design programs.
3. To study and apply different types of java concepts like multithreading ,packages ,exception handling ,interfaces and design programs using these concepts.
4. To study and apply various features of AWT components.
5. To know the basic concepts of networking in the context of java programming.

Course Outcomes:

1. Ability to define different procedural and object oriented concepts and will be able to differentiate between them.
2. Ability to define, understand and differentiate different types of arrays and apply them.
3. Ability to recognize various concepts of java and develops the programs using them.
4. Ability to identify and differentiate the various features of AWT components to construct container based programs.
5. Ability to describe and explain the concept of networking.

SYLLABUS

UNIT-I:

Overview, Environment Set Up, Basic Syntax, Identifiers, Reserved words, Data Types, Literals, Basic Operators

UNIT-II:

Control Statements in Java: if...else statement, for, while, do-while, for-each, Nested for loops, switch, break, continue, return, Objects & Classes, Access Specifiers, Input & Output, Arrays, Strings

UNIT-III:

Methods, Relationship between objects, Object-Oriented Programming: Encapsulation, Abstraction, Inheritance, Polymorphism, Interfaces, Type Casting, Packages

UNIT-IV:

Exception Handling: try, catch, final, finally, throw, throws, Built-in, User-defined Exceptions,
Files: Read, Write and Append operations using text streams & byte streams

UNIT-V:

Collection Framework, Generics

UNIT-VI:

Threads: life cycle, single tasking, multi tasking, Deadlocks, Thread Priorities, Daemon Threads,
Serialization

References:

1. JAVA Complete Reference by Herbert Schildt
2. Core JAVA An Integrated Approach by Dr. R. Nageswara Rao.
3. <http://spoken-tutorial.org/>

ENGLISH PROFICIENCY-II

(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

To equip the students with the components of the language required and help them gain adequate knowledge so as to become employable and competent.

Course Objectives:

Students will be able to

1. Enhance their Interpretative skills
2. Understand how to prepare a text.
3. Comprehend various types of writing discourses and respond.
4. Produce effectively different write-ups related to various business contexts.
5. Strengthen their emotional make-up.
6. Perceive various writing discourses.

Course Outcomes:

The students will

1. Develop the skills of taking and making notes
2. Interpret the pictures appropriately and effectively.
3. Read, comprehend and infer a given piece of writing effectively.
4. Learn and practice the skills of Research writing.
5. Communicate well through various forms of writing.
6. Be confident in giving presentations and dealing with people.

SYLLABUS**UNIT-1:SPEAKING**

Analyzing proverbs

Enactment of One-act play

UNIT-2:READING

Reading Comprehension

Summarizing Newspaper Article

UNIT-3:WRITING

Note Taking & Note Making

Precis Writing

Essay Writing

Letter Writing

Picture Description

Literary Appreciation– Learning the Language of Literature

UNIT-4: VOCABULARY

Indian-origin English Words

Phrasal Verbs for Day-to-Day Communication

Commonly used Idiomatic Expressions

UNIT-5: PROJECT

Research Writing

Reference Books:

1. English for Colleges by Brendan J. Carroll, Macmillan Publications
2. Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw Hill
3. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
4. The Oxford Guide to Writing and Speaking by John Seely OUP
5. Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
6. English for Engineers and Scientists by Sangeeta Sharma and Binod Mishra, PHI Learning



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

Under Choice Based Credit System

ELECTRICAL & ELECTRONICS ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/Week	Internal Marks	External Marks	Total Marks
B17 EE 3101	Electrical Machines-II	3	3	1	--	4	30	70	100
B17 EE 3102	Microprocessor & Microcontroller	3	3	1	--	4	30	70	100
B17 EE 3103	Electrical Power Generation, Transmission & Distribution	3	3	1	--	4	30	70	100
B17 EE 3104	Control Systems	3	3	1	--	4	30	70	100
B17 EE 3105	Digital Electronics & Logic Design	3	3	1	--	4	30	70	100
B17 EE 3106	Non-Conventional Energy Sources	3	3	1	--	4	30	70	100
B17 EE 3107	Electrical Machines Lab-I	2	--	--	3	3	50	50	100
B17 EE 3108	Microprocessor & Microcontroller Lab	2	--	--	3	3	50	50	100
B17 BS 3101	Problem Solving & Linguistic Competence	1	--	3	--	3	30	70	100
B17 BS 3102	Basic Coding	1	--	--	3	3	50	50	100
B17 BS 3105	IPR & Patents	--	--	2	--	2	--	---	--
Total		24	18	11	9	38	360	640	1000

ELECTRICAL MACHINES-II

Lecture : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives: Students will

1. Understand the concept of principle of operation of 3 phase induction motor, equivalent circuit slip torque characteristics, and circle diagram.
2. Understand the single phase induction motor, double field revolving theory, and equivalent circuit capacitor start motor.
3. Understand the operation and the principle of synchronous machine, voltage regulation, power angle characteristics synchronizing power, synchronizing torque.
4. Understand different starting methods of synchronous motor, effect of damper windings and Hunting.
5. Understand design procedures of synchronous machine and induction machines.

Course Outcomes:

Students are able to

1. Understand the concepts of construction, operating principle and starting methods of AC machines.
2. Perform various tests on AC Machines
3. Analyze the performance of different AC machines in the concepts of torque and power factor correction.

SYLLABUS**UNIT-I BASIC CONCEPTS OF ELECTRICAL MACHINES:**

Physical arrangement of windings in stator and cylindrical rotor; slots for windings; single turn coil - active portion and overhang; full-pitch coils, concentrated winding, distributed winding, winding axis, generated emf, Air-gap MMF distribution with fixed current through winding - concentrated and distributed, Sinusoidally distributed winding, winding distribution factor

UNIT-II INDUCTION MACHINES

Construction, Types (squirrel cage and slip-ring), Equivalent circuit. Phasor Diagram, Torque Slip Characteristics, Starting and Maximum Torque. Losses and Efficiency. Circle Diagram. Effect of parameter variation on torque speed characteristics (variation of rotor resistance). Methods of starting, braking and speed control for induction motors. Generator operation. Self-excitation. Doubly-Fed Induction Machines.

UNIT-III SINGLE PHASE INDUCTION MOTORS

Constructional features, double revolving field theory, equivalent circuit, determination of parameters. Split phase starting methods & applications.

UNIT-IV SYNCHRONOUS GENERATORS

Constructional features, Cylindrical rotor machines, Synchronous Generator-circuit model and phasor diagram, armature reaction, synchronous impedance, voltage regulation and estimation of voltage regulation by EMF, MMF and ZPF methods, Salient pole Machine-Two reaction theory, analysis of phasor diagram, power angle characteristics, determination of x_d and x_q , Parallel operation of Alternators-Synchronization and load division.

UNIT-V SYNCHRONOUS MOTORS

Operating principle, circuit model, phasor diagram, effect of load, Operating characteristics of synchronous machines, V-curves, starting methods of synchronous motors.

Text Books:

1. Nagrath & Kothari, "Electric Machines" TMH 5th edition
2. PS Bhimbra, "Electrical Machinery", Khanna Publishers. 7th edition.

Reference Books:

1. Fitzgerald & Kingsley, "Electric Machinery" McGraw Hill, 6 edition
2. Alexander S. Langsdorf, "AC Machines", Tata McGraw Hill. 2nd edition

MICROPROCESSOR & MICROCONTROLLER

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives

Students will

1. Study the architecture of 8085 microprocessor, functions, instruction sets etc.
2. Architecture of different peripherals like 8259,8255,8279 and A/D and D/A convertors
3. Study the architecture of 8051 microcontroller, functions etc.
4. Study 8086 architecture and addressing.

Course Outcomes

Students are able to

1. Understand the fundamentals of 8085 Microprocessor and microcontroller based systems.
2. Familiarize with the instruction set and assembly level programming.
3. Illustrate how the different peripherals (8255, 8253 etc.) are interfaced with Microprocessor.
4. Distinguish and analyze the properties of Microprocessors & Microcontrollers.
5. Apply knowledge on interfacing microcontrollers for some real time applications.

SYLLABUS**UNIT-I 8085 MICROPROCESSOR:**

Introduction to microprocessors, micro computers – Architecture of 8085 microprocessor – pin-out diagram of 8085 – Detailed description of the 8085 pins – addressing modes Memory interfacing – Machine cycles and bus timings for Opcode fetch, memory read, memory write, I/O read, I/O write operations – Memory mapped I/O and I/O mapped I/O.

UNIT-II 8085 INSTRUCTIONS AND PROGRAMMING:

Difference between Machine language, Assembly language and High level language – Brief description of the 8085 instruction set – 8085 programming using data transfer group, arithmetic group, logical group, branch transfer group, stack and subroutines – counters and delay .

UNIT-III INTERFACING PERIPHERALS TO 8085:

Function of D/A and A/D converters – Interfacing D/A and A/D converters. Detailed description and interfacing of 8251 USART, 8253/8254 programmable timer, 8255 PPI, 8257 DMA controller, 8279 programmable keyboard/display interface

UNIT-IV 8051 MICROCONTROLLER:

Introduction to microcontrollers – Comparison between microprocessors and microcontrollers – Functional block diagram of 8051 microcontroller and its description – 8051 pin-out diagram and description of 8051 pins – Interfacing external memory to 8051 – implementing counters and timers in 8051 – Serial data transfer using 8051 – Various interrupts and its programming in 8051. Interfacing Stepper motor to 8051 microcontroller.

UNIT-V ADVANCED TOPICS IN MICROPROCESSORS:

Architecture of 8086 microprocessor pin out diagram – Addressing modes – differences between 8085 and 8086.

Text Books:

1. Ramesh S. Gaonkar, Microprocessor Architecture, Programming, and Applications with the 8085, Penram International Publishing, New Delhi, 5th edition, 2008.
2. Kenneth. J.Ayala, The 8051 Microcontroller, Cengage learning, 3rd edition ,2005.
3. A.K.Ray and Bhurchandi, “Advanced Microprocessors and Peripherals”, 2nd Edition, TMH Publications

Reference Books:

1. Douglas V.Hall, “Microprocessors and Interfacing”, 2nd Revised Edition, TMH Publications.Design”, 2nd ed., PHI
2. Krishna Kant, “Microprocessors and Microcontrollers”, PHI Publisher.

ELECTRICAL POWER GENERATION, TRANSMISSION & DISTRIBUTION

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students will

1. Understand the general arrangement, principles and components & their functions present in thermal, hydro, nuclear and gas power plants.
2. Understand the load curves and different types of tariffs.
3. Know the performance analysis of transmission lines
4. Give emphasis on mechanical design of transmission line cables and insulators.
5. To study the different types of distribution systems.

Course Outcomes

After completion of the course student will be able to

1. Describe the power generation from different energy sources.
2. Demonstrate different tariffs of generation, Inductance & Capacitance of transmission lines.
3. Analyze the various transmission and distribution systems under various conditions.

SYLLABUS**UNIT-I ELECTRIC POWER GENERATION& ECONOMIC CONSIDERATIONS:**

Layout of thermal, hydro, nuclear and gas power plants, brief description of various parts of different power plants. Load curves and associated definitions, load duration curves, different types of tariffs and examples.

UNIT-II POWER SUPPLY SYSTEMS& DISTRIBUTION SYSTEMS:

Transmission and distribution systems- D.C 2-wire and 3- wire systems, A.C single phase, three phase and 4-wire systems, comparison of copper efficiency. Primary and secondary distribution systems, concentrated & uniformly distributed loads on distributors fed at both ends, ring distributor, voltage drop and power loss calculation, Kelvin's law.

UNIT-III INDUCTANCE & CAPACITANCE CALCULATIONS:

Types of conductors, line parameters, calculation of inductance and capacitance of single and double circuit transmission lines, three phase lines with bundle conductors. Skin effect and proximity effect.

UNIT-IV PERFORMANCE OF TRANSMISSION LINES:

Generalized network constants and equivalent circuits of short, medium, long transmission line. Line performance: regulation and efficiency, Ferranti effect.

UNIT-V OVERHEAD LINE INSULATORS:

Types of insulators, potential distribution over a string of suspended insulators, methods of equalizing potential. Corona: phenomenon of corona, corona loss, concept of radio interference.

MECHANICAL DESIGN OF TRANSMISSION LINES :

Different types of tower, sag –tension calculations, sag template, string charts.

Text Books:

1. Wadhwa,C.L., “ Electric Power Systems”, New Age International Private Limited; Sixth edition (2010).
2. Power System Analysis and Design by Dr. B.R Gupta S Chand &Company; 2005.
3. Nagarath,I.J, and Kothari, D.P., “Power System Engineering”, McGraw Hill Education; 2 edition (2007).
4. “A Course in PowerSytems” by J.B Gupta, S.K. Kataria& Sons; 2013 edition.
5. “ Principles of power systems” by V.k Mehta & Rohit Mehta by S.CHAND Publications, 3rd edition 2005.

Reference Books:

1. Burke James,J., “Power Distribution Engineering; Fundamentals and Applications “ Marcel Dekker 1996.
2. Grainger john, J. and Stevenson ,Jr. W.D., “Power System Analysis”, McGraw Hill,1994.
3. Harder Edwin,I., “Fundamentals of Energy Production”,JohnWiely and Sons,1982.
4. Deshpande, M.V., “Elements of Electric Power Station Design”,A.H Wheeler andCo.Allahabad,1979.

CONTROL SYSTEMS

Lecture : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course objectives

Students will

1. Learn the modelling of linear systems using transfer functions and obtain transfer functions for physical electrical and mechanical systems.
2. Learn to represent systems using block diagrams and signal flow graphs and derive their transfer functions.
3. Learn the significance of time response and find it for system analysis in transient and steady state.
4. Learn the concept of stability and know different techniques of stability analysis.
5. Learn the concept of frequency response and its application for control system analysis.

Course Outcomes

Students are able to

1. Model electrical and mechanical physical systems by applying laws of physics
2. Represent mathematical models of systems using block diagrams & Signal Flow Graphs and derive their transfer functions
3. Analyze systems in time domain for transient and steady-state behavior
4. Learn the concept of stability and use RH criterion and Root locus methods for stability analysis.
5. Learn to obtain frequency response plots of systems and use them for system analysis and stability assessment.

SYLLABUS**UNIT I:**

Introduction to control systems- Open loop and closed loop systems- Transfer Functions of Linear Systems– Impulse Response of Linear Systems – Mathematical Modeling of Physical Systems – Equations of Electrical Networks – Modeling of Mechanical Systems – Equations of Mechanical Systems, Analogous Systems.

UNIT II:

Block Diagrams of Control Systems – Signal Flow Graphs (Simple Problems) – Reduction Techniques for Complex Block Diagrams and Signal Flow Graphs (Simple Examples)- Feedback Characteristics of Control Systems

UNIT III:

Time Domain Analysis of Control Systems – Time Response of First and Second Order Systems with Standard Input Signals – Steady State Error Constants – Effect of Derivative and Integral Control on Transient and Steady State Performance of Feedback Control Systems.

UNIT IV:

Concept of Stability– Routh-Hurwitz Criterion, Relative Stability Analysis, the Concept and Construction of Root Loci, Analysis of Control Systems with Root Locus (Simple Problems to understand theory).

UNIT V:

Frequency Domain Analysis of control systems - Bode Plots- Log Magnitude versus Phase Plots- Polar Plots -Correlation between Time and Frequency Responses - Nyquist Stability Criterion -Assessment of Relative Stability -All Pass and Minimum Phase Systems - Constant M and N Circles.

Text Books:

1. I. J. Nagrath and M. Gopal, 'Control Systems Engineering', New Age International Publishers(6th Edition).

Reference Books:

1. Benjamin C. Kuo, 'Automatic Control Systems' , PHI (5th Edition).
2. Katsuhiko Ogata, 'Modern Control Engineering' , , PHI (4th Edition).
3. Richard C. Dorf and Robert H. Bishop, 'Modern Control Systems', Addison-Wesley Publishers(8th Edition)

DIGITAL ELECTRONICS AND LOGIC DESIGN

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students will

1. Study the basic philosophy underlying the various number systems, negative number representation, binary arithmetic, binary codes and error detecting and correcting binary codes.
2. Study the theory of Boolean algebra and to study representation of switching functions using Boolean expressions and their minimization techniques.
3. Study the combinational logic design of various logic and switching devices and their realization.
4. Study the sequential logic circuits design both in synchronous and Asynchronous modes for various complex logic and switching devices, their minimization techniques and their realizations.
5. Study some of the programmable logic devices and their use in realization of switching functions.

Course Outcomes:

Students are able to

1. Understand the concepts of basic number system and Boolean
2. Apply the Boolean algebra for framing the simplified expression.
3. Analyze the combinational & sequential circuits using simple logic gates and PLD & PLA.

SYLLABUS

UNIT I NUMBERING SYSTEMS: Digital systems - Binary, Octal, Decimal and Hex numbering systems – Number base Conversions – $(n-1)$'s and n 's complements of the various numbering systems – Binary arithmetic – Various methods to represent signed binary numbers. Binary Codes: BCD, Excess-3 codes – Binary arithmetic using BCD and Excess-3 codes – Gray code – Error detecting codes: parity checking and Hamming code – Error correcting codes: Hamming code

UNIT II BOOLEAN ALGEBRA AND BOOLEAN FUNCTIONS: Boolean theorems and postulates – Logic gates – Truth table - Boolean functions – Dual of a function – Complement of a function – Canonical and standard forms – Simplification of Boolean functions using Boolean theorems and postulated, Karnaugh map (K-map) with maximum of 4 variables

UNIT III COMBINATIONAL LOGIC CIRCUITS: Boolean function implementation using AND-OR logic, multilevel NAND and multilevel NOR implementation – Transformation of multilevel NAND and NOR circuits to AND-OR diagram – Combinational logic design - Half adder – Full adder – Half subtractor – Full subtractor – Parallel adder – Parallel adder/subtractor – Carry look ahead adder – BCD adder – Magnitude comparator – code converters, Decoders – Encoders – Demultiplexer – Multiplexer – Logic implementation using Programmable Logic Devices.

UNIT IV SEQUENTIAL LOGIC CIRCUITS: Differences between combinational logic and sequential logic – Flip-flops (R-S, J-K, D, T, Master-slave J-K flip-flop) – Truth tables and excitation tables of the flip-flops, Conversions of flip-flops. Digital Counters-Ripple Counter design, Synchronous Counter design with T, D and J.K. Flip-flops. Shift Registers and Operation Modes.

UNIT V REALIZATION OF LOGIC GATES USING DIODES & TRANSISTORS: AND, OR and NOT Gates using Diodes and Transistors, RTL, DTL, TTL and CML Logic Families and its Comparison.

Text Book:

1. M. Morris Mano, Digital Design, Prentice-Hall of India Pvt. Limited, New Delhi, 2nd Edition. 2000.

Reference Books:

1. ZviKohavi, Switching and Finite Automata Theory, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2nd Edition, 2008.
2. Frederick J. Hill and Gerald R. Peterson, Introduction to Switching Theory and Logic Design, John Wiley & sons, Inc. New York, 3rd edition, 1981.

NON-CONVENTIONAL ENERGY SOURCES

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To know the need for Renewable energy.
2. To understand the concepts of solar energy and its conversion systems.
3. To Understand the basic physics of wind power generation.
4. To be familiar with the ocean, wave and tide energies.
5. To understand the concepts of geo-thermal energy and bio energy.

Course Outcomes:

After doing this course, the students will be able to

1. Identify the need for Renewable energy
2. Recognise the ways of collection of solar energy.
3. Apply the knowledge of wind energy to estimate the energy potential.
4. Apply the knowledge of ocean, waves and tides to estimate their energy potential.
5. Understand the concepts behind geo-thermal energy and bio energy.

SYLLABUS**UNIT-I:****Introduction to Non-Conventional Energy Sources:**

Environmental aspects of conventional electric energy generation, renewable and non-conventional energy sources, impact of renewable energy generation on environment, prospects of renewable energy sources.

UNIT-II:**Solar Energy:**

Solar radiation and its measurements: introduction to solar energy, solar constant, solar radiation at the earth's surface, solar radiation geometry, solar radiation measurements, estimation of average solar radiation, solar radiation on tilted surface. Solar energy collectors: physical principles of the conversion of solar radiation into heat, flat plate collectors, concentrating collectors, advantages and disadvantages. Solar electric power generation: principles of solar photo-voltaic cells, conversion efficiency and power output.

UNIT-III:**Wind Energy:**

Introduction, basic principles of wind energy conversion-nature of wind, power in the wind, maximum power, forces on the blades, lift and drag forces, aerodynamics, types of wind power plants, types of wind turbine - generating units, generating systems, energy storage, application of wind energy, site selection considerations, environmental aspects.

UNIT-IV:

Ocean Energy: Ocean thermal energy conversion: working principle, availability, types, advantages, limitations and applications. Wave energy: factors affecting the wave energy, mathematical analysis for potential energy, kinetic energy, total energy and wave power. Tidal energy: basic terminology, types of tidal plants, energy potential estimation from a tidal plant, advantages and limitations.

UNIT-V:

Geo-Thermal Energy: Structure of earth's interior, thermal gradient, geo-thermal energy sources, types of geo-thermal power generation, merits, demerits and applications of geo-thermal energy.

Bio Energy: Overview, bio-mass conversion processes, bio-gas generation, factors affecting the generation of bio gas, various types of bio gas plants.

Text Books:

1. G. D. Rai, "Non-Conventional Energy Sources", fifth edition, Khanna Publishers, 2015.
2. D. P. Kothari, K. C. Singal and Rakesh Ranjan, "Renewable Energy sources and Emerging Technologies", 2nd Edition, PHI Learning Pvt. Limited, 2013.
3. S. Hasan Saeed and D. K. Sharma, "Non-Conventional Energy Resources", First Edition, S.K. Kataria & Sons, 2013.

Reference Books:

1. S. P. Sukhatme, "Solar Energy", Third edition, Tata McGraw-Hill Education, 1996.
2. S. Sumathi, L. Ashok Kumar and P. Surekha, "Solar PV and Wind Energy Conversion Systems (Green Energy and Technology)", First Edition, Springer, 2015
3. G. N. Tiwari and M. K. Ghosal, "Renewable energy resources", First Edition, Narosa Publishing House, 2004.

ELECTRICAL MACHINES-I LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int. Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

Students will

1. Conduct experiments on characteristics of generators & motors
2. Perform Load tests on series, shunt, compound motors and compound generators-Swinburne's, Hopkinson's test.
3. Conduct OC & SC tests on single phase transformers, Sumpner's test.

Course Outcomes:

Students are able to

1. analyze characteristics of various types of generators & motors which will help in Understanding of machines under various conditions.
2. Compare Speed control of dc motors which will be useful in various industries.
3. Determine testing of machines will give an idea in testing side in various industries.

LIST OF EXPERIMENTS

1. Swinburne's Test
2. Speed control of a DC shunt motor.
3. Load test on DC Shunt motor.
4. Load test on DC series motor.
5. Load test on DC Compound generator.
6. Open circuit characteristics of a DC shunt generator.
7. Hopkinson's Test.
8. Internal and external characteristics of a DC shunt generator.
9. OC and SC tests on a single phase transformer.
10. Load test on a single phase transformer.
11. Sumpner's Test.

Text Books:

1. Kothari.D.P and Nagrath.I.J., "Electrical Machines", Tata McGraw Hill Publishing Co.Ltd, New Delhi, 5th edition 2002.
2. Bimbhra.P.S, Electrical Machinery, Khanna Publishers, IL Kosow, "Electrical Machines & Transformers", Prentice Hall of India. 2nd edition 2003.

Reference Books:

1. Fitzgerald, A.E., Charles Kingsely Jr. Stephen D. Umans, "Electric Machinery" McGraw Hill Books Company, 6th edition 2002.
2. Hill Stephen, Chapman.J, "Electric Machinery Fundamentals", McGraw Hill Book Co., New Delhi, 4th edition 2005.

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MICROPROCESSOR AND MICRO CONTROLLER LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int. Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

Students will

1. Learn about working of 8085 microprocessor and instructions sets.
2. Acquire knowledge of writing program for arithmetic, logical operation to perform required task like pick largest even number, 8-bit array addition etc.
3. Understand the programs to display decimal count & converting binary to BCD codes and also learns about sorting techniques.
4. Learn about working of 8051 microcontroller and instructions sets
5. Learn about basic programming skills using 8051 controller and also acquire knowledge on interfacing with other devices like traffic light controller, stepper motor etc

Course Outcomes:

Students are able to

1. Evaluate the programs using basic fundamentals of 8085 Microprocessor& 8051 Microcontroller.
2. Develop different programs on extended version like 8086 microprocessor.
3. Design programs for interfacing circuits like traffic controller, LED display board, Motor controllers etc.
4. Utilize their knowledge practically in PLC designs companies. Ex: Govt.sector& Private sectors

LIST OF EXPERIMENTS**PART A: Experiments on Microprocessors:**

1. Program to add two 8-bit binary numbers
2. Program to add an array of 8-bit binary numbers.
3. Program to pick the largest even number from an array of 8- bit binary numbers
4. Program to find the sum of an array of 2- digit packed BCD numbers.
5. Program to display decimal count from 0 to 9 with suitable delay between each count.
6. Program to convert an 8- bit binary number into BCD.
7. Program to sort given array of 8-bit binary numbers.

PART B: Experiments on Micro Controllers:

8. Microcontroller programming on two 8-bit numbers
a) Addition, b) Subtraction, c) Multiplication & d) Division
9. Program to obtain decimal equivalent of an 8-bit hexadecimal number
10. Interfacing stepper motor and speed control using 8051 microcontroller
11. Traffic light control using 8051 microcontroller.

Text Books:

1. R.S. Gaonkar: Microprocessor Architecture.
2. Microprocessors & Its Applications By Theagarajan, R., Dhanpal, S. & Dhanaseturan, S., New Age India Ltd., 1998
3. A.K. Ray and Bhurchandi, "Advanced Microprocessors and Peripherals", 2nd Edition, TMH Publications
4. Ajay V. Deshmukh, "Microcontrollers, Theory and applications", Tata McGraw-Hill Companies – 2005.

Reference Books:

1. Douglas V. Hall, "Microprocessors and Interfacing", 2nd Revised Edition, TMH Publications.
Design", 2nd ed., PHI
2. Kenneth. J. Ayala, The 8051 Microcontroller Architecture, Programming and Applications, Penram International 2nd edition, 1996.
3. Krishna Kant, "Microprocessors and Microcontrollers", PHI Publishers

PROBLEM SOLVING & LINGUISTIC COMPETENCE
(Common to all Branches)

Theory	: 3 Periods (VA-2+QA-1)	Int.Marks	: 30
Exam	: 3 Hrs.	Ext.Marks	: 70
		Credits	: 1

Part-A: Verbal and Soft Skills-I

Course Objectives:

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS

Grammar: (VA)

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis – Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners' Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cambridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

BASIC CODING
(Common to ECE & EEE)

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 1

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:**At the end of the course students will be able to**

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS

UNIT I Review of Programming constructs

Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers

Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes

Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.

Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations vs pointed locations, Referencing a 2D-Matrix

UNIT IV User-defined datatypes, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by RobotLaphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>

IPR & PATENTS
(Common to CE, EEE & ME)

Tutorial : 2 Periods

Credits : 0

Course Objectives:

1. To introduce the idea of tangible and intangible property and its protection.
2. To familiarize with the frameworks for protection of intellectual property.
3. To layout the procedures to claim rights over intellectual property.

Course Outcomes:

After successful completion of the course, the student shall be able to

1. Identify various types of intangible property that an engineering professional could generate in the course of his career.
2. Distinguish between various types of protection granted to Intellectual Property such as Patents, Copy Rights, Trademarks etc.,
3. List the steps involved in getting protection over various types of intellectual property and maintaining them.
4. Take precautions in writing scientific and technical reports without plagiarism.
5. Help micro, small and medium entrepreneurs in protecting their IP and respecting others IP as part of their business processes.

SYLLABUS

UNIT I

Intellectual Property Law: Basics - Types of Intellectual Property - Innovations and Inventions - Trade related Intellectual Property Rights – Agencies Responsible for Intellectual Property Registration – Infringement - Compliance and Liability Issues

UNIT II

Principles of Copyright – Subject Matters of Copyright – Rights Afforded by Copyright Law – Copyright Ownership – Copyright Formalities and Registration – Limitations – Infringement of Copyright - Plagiarism and difference between Copyright infringement and Plagiarism

UNIT III

Introduction to Trade Mark – Trade Mark Registration Process – Post registration procedures – Trade Mark maintenance– Infringement – Dilution of Ownership of Trade Mark – Likelihood of confusion – Trade Mark claims – Trade Marks Litigation – International Trade Mark Law

UNIT IV

Introduction to Patent Law – Rights and Limitations – Rights under Patent Law – Patent Requirements – Ownership and Transfer – Patent Application Process and Granting of Patent – Patent Infringement and Litigation – International Patent Law – Double Patenting

UNIT V

Introduction to Trade Secrets – Maintaining Trade Secret – Physical Security – Employee Access Limitation – Employee Confidentiality Agreement – Trade Secret Law – Unfair Competition – Trade Secret Litigation – Breach of Contract – Applying State Law.

Text Books:

1. Kompal Bansal & Parikshit Bansal "Fundamentals of Intellectual Property for Engineers", BS Publications
2. Prabhuddha Ganguli: "Intellectual Property Rights" Tata McGraw –Hill, New Delhi
3. R. Radha Krishnan, S. Balasubramanian: "Intellectual Property Rights: Text and Cases", Excel Books, New Delhi.

Reference Books:

1. Deborah E. Bouchoux: "Intellectual Property". Cengage learning , New Delhi
2. Richard Stim: "Intellectual Property", Cengage Learning, New Delhi.

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

III/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)
Under Choice Based Credit System

ELECTRICAL & ELECTRONICS ENGINEERING
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 EE 3201	Advanced Control Systems	3	3	1	--	4	30	70	100
B17 EE 3202	Power Electronics	3	3	1	--	4	30	70	100
B17 EE 3203	Power System Analysis And Stability	3	3	1	--	4	30	70	100
B17 EE 3204	Digital Signal Processing	3	3	1	--	4	30	70	100
B17 EE 3205	Power System Protection	3	3	1	--	4	30	70	100
# OE	Open Elective	3	3	1	--	4	30	70	100
B17 EE 3207	Electrical Machines –II Lab	2	--	--	3	3	50	50	100
B17 EE 3208	Control Systems Lab	2	--	--	3	3	50	50	100
B17 BS 3201	Employability Skills	1	--	3	--	3	30	70	100
B17 BS 3203	Advanced Coding	1	--	--	3	3	50	50	100
Total		24	18	9	9	36	360	640	1000

OPEN ELECTIVE	B17 CS 3214	Oops Through Java
	B17 CS 3217	Unix & Shell Programming
	B17 CS 3218	Neural Networks & Fuzzy Logic
	B17 EC 3203	VLSI Design
	B17 ME 3210	Industrial Robotics

ADVANCED CONTROL SYSTEMS

Lecture : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

Students will

1. Understand control system components construction and application.
2. Study state variable techniques.
3. Learn Z-transforms and inverse Z-transform.
4. Understand classical and modern controllers.
5. Study stability tests- Bilinear transformation and Jury's test.

Course Outcomes:

Students are able to:

1. Know the various components and usage of each component.
2. Derive state space model for a given systems and Apply the concept of Observability and Controllability for LTI system.
3. Apply Z- transform in Engineering application related to digital control systems.
4. Design classical controller based on bode plots and modern controllers based on the state space techniques
5. Test the digital system which is useful after designing a particular system with respect to the stability point of view.

SYLLABUS

UNIT I CONTROL SYSTEMS COMPONENTS: D.C. & A.C. Tachometers- Synchros, A.C. and D.C. Servo Motors-Stepper Motors and Its Use in Control Systems, Amplidyne, Metadyne, Magnetic Amplifier –Principle, Operation.

UNIT II STATE VARIABLE ANALYSIS: Concept of State, State Variables & State Models, State Model for Linear Continuous Time Systems, Solution of State Equation, State Transition Matrix, Concept Of Controllability & Observability (Simple Problems To Understand Theory).

UNIT III THE Z-TRANSFORM: Introduction To Z-Transforms and Inverse Z-Transforms. (Simple Problems to Understand Theory).

UNIT IV INTRODUCTION TO DESIGN: Introduction-Preliminary Considerations of Classical Design-Lead Compensation-Lag Compensation-Realization of Compensating Networks-Cascade Compensation in Frequency Domain (Bode Plot Techniques) - Pole Placement By State Feed-Back.

UNIT V STABILITY: Stability of Linear Digital Control Systems, Definition & Theorem, Stability Tests, Bi-Linear Transformation Method, Jury's Stability Test.

Text Books:

1. Gibson, John E.: Control System Components" to "Gide, André, 1869-1951
2. Automatic Control System (U.P.) by R.C.Shukla
3. Control Systems Engineering. by I.J. Nagrath, M.Gopal Vth edition
4. Automatic Control Systems, 9ed Paperback – 2014 Benjamin C. Kuo Farid Golnaraghi Ninth Edition.

Reference Books:

1. Digital Control System Second Edition Benjamin C. Kuo second edition
2. Control Systems: Principles and Design M. Gopal Fourth edition Tata McGraw-Hill Education, 2002.
3. Digital control systems. Theory, hardware, software constantine H. Houpis and Gary B. Lamont McGraw-Hill 1985 second edition
4. Discrete - Time Control Systems Paperback – 2005 second edition

POWER ELECTRONICS

Lecture : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

Students will

1. Understand the concepts of Power Semiconductor devices and their applications.
2. Understand the need of Energy conversion and effective implementation methods.

Course outcomes:

Students are able to

1. Explain the principle of operation of thyristor, modern power semiconductor devices and necessity of series and parallel connection of thyristors.
2. Explain the operation of Firing and Commutation techniques.
3. Evaluate the phase controlled rectifiers with different loads.
4. Analyze different Choppers, Cyclo-converter and AC voltage Controller configurations.
5. Investigate harmonic reduction techniques for inverters based on PWM techniques.

SYLLABUS**UNIT I MODERN POWER SEMI CONDUCTOR DEVICES**

Thyristors – Silicon Controlled Rectifiers (SCRs) – BJT – Power MOSFET – Power IGBT and their characteristics. Basic theory of operation of SCR – Static characteristics and Dynamic characteristics of SCR - Turn on and Turn off times – Turn on and turn off methods. Two transistor analogy of SCR -Series and parallel connections of SCRs Snubber circuit details – Numerical problems.

UNIT II THYRISTOR FIRING AND COMMUTATION CIRCUITS

SCR trigger circuits-R, RC and UJT triggering circuits. The various commutation methods of SCRs-Load commutation- Resonant Pulse Commutation- Complementary Commutation- Impulse Commutation- External Pulse Commutation Techniques. Protection of SCRs

UNIT III PHASE CONTROLLED RECTIFIERS

Principles of phase controlled rectification -Study of Single phase and three-phase half controlled and full controlled bridge rectifiers with R, RL, RLE loads. Effect of source inductance. Dual converters- circulating current mode and circulating current free mode-control strategies. Numerical problems.

UNIT IV CHOPPERS, CYCLO CONVERTER AND AC VOLTAGE CONTROLLER

Classification of Choppers A, B, C, D and E, Switching mode regulators-Study of Buck, Boost and Buck-Boost regulators, Cuk regulators . Principle of operation of Single phase bridge type Cycloconverter and their applications. Single phase AC Voltage Controllers with R and RL loads.

UNIT V INVERTERS

Principle of operation of Single phase Inverters -Three phase bridge Inverters (180° and 120° modes)-voltage control of inverters-Single pulse width modulation- multiple pulse width modulation, sinusoidal pulse width modulation. Harmonic reduction techniques- Comparison of Voltage Source Inverters and Current source Inverters.

Text Books:

1. Power electronics - P.S. Bimbhra- Khanna Publishers, 4th Edition
2. Power electronics – M.D. Singh & K.B. Kanchandhani, Tata Mc Graw – Hill Publishing Company, 2nd edition.

Reference Books:

1. Power Electronics: Circuits Devices and Applications – M.H. Rashid, Prentice Hall of India, 3rd edition.
2. Power Electronics – VedamSubramanyam, New Age International (p) Limited, Publishers.
3. Power Electronics – P.C. Sen, Tata Mc Graw-Hill Publishing.
4. Thyristorised power Controllers – G.K. Dubey, S.R Doradra, A. Joshi and R.M.K. Sinha, New Age international Pvt Ltd. Publishers latest edition

POWER SYSTEM ANALYSIS AND STABILITY

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives

Students will

1. Acquire knowledge for the design and analysis of electrical power grids.
2. Learn the modelling of single line diagram and formation of Y-bus
3. Study the calculation of power flow in a power system network using various load flow techniques,
4. Learn about different types of power system faults and analysis.
5. Study the stability of system under steady state and transient period.

Course Outcomes

Students are able to

1. Understand single line diagram, reactance diagram of the power system.
2. Apply different load flow techniques to solve power system problems.
3. Analyse different types of faults in a power system
4. Analyse steady state and transient stability problems of power system.

SYLLABUS

UNIT I P.U. REPRESENTATION: Single Line Diagram, Per Unit Quantities, P.U. Impedance of 3-Winding Transformers, P.U. Impedance Diagram of a Power System.

UNIT II LOAD FLOW STUDIES: Formulation of Network Matrices, Load Flow Problem, Gauss-Seidel Method, Newton-Raphson Method & Fast Decoupled Method of Solving Load Flow Problem.

UNIT III SYMMETRICAL FAULT ANALYSIS: 3-Phase Short Circuit Currents and Reactance's of a Synchronous Machine, Fault Limiting Reactors.

UNIT IV SYMMETRICAL COMPONENTS: The Symmetrical Components, Sequence Impedances and Sequence Networks. Phase Shift in Delta/Star Transformers, 3-Phase Power in terms of Symmetrical Components.

UN-SYMMETRICAL FAULTS: LG, LL and LLG Faults on an Unloaded Alternator.

UNIT V POWER SYSTEM STABILITY: Concepts of Stability (Steady State And Transient), Swing Equation, Equal Area Criterion, Critical Clearing Angle and Time for Transient Stability, Step by Step Method of Solution, Factors Affecting Transient Stability.

Text Books:

1. Power System Engineering by J.G. Nagarith& D.P. Kothari,4th Edition TMH Publication
2. Elements of Power System Analysis, William D. Stevenson, Jr, Mc Graw Hill Pub first edition, 1955.
3. Power System Analysis And Design By Dr. B.R.Guptha, S Chand & Company, Aug 2005.

Reference Books:

1. Power System Analysis by Hadi Sadat, Mc Graw Hill Pub 1999.
2. Power System Stability & Control by Prabha Kundur, TMH Publication, JAN 22, 1994.

DIGITAL SIGNAL PROCESSING

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

This course will introduce the basic concepts and techniques for processing signals. By the end of the course, student will be familiar with the most important methods in DSP, including digital filter design, transform-domain processing and importance of Signal Processors.

Course Outcomes:

Students are able to

1. Analyse difference equations of linear time-invariant systems and Evaluate the transfer functions using Z transforms.
2. Represent signals mathematically in continuous and discrete-time, and in the frequency domain.
3. Solve the linear and circular convolutions of discrete-time sequences.
4. Understand the Discrete-Fourier Transform (DFT) and the FFT algorithms, relate it to the DTFT.
5. Design IIR& FIR filters.

SYLLABUS**UNIT-I DISCRETE - TIME SIGNALS AND SYSTEMS:**

Discrete Time Signals & Sequences, Z-Transform and ROC, Linear Shift – Invariant Systems, Stability And Causality, Linear Constant Coefficient Difference Equations, System Function $H(Z)$ Of Digital Systems, Structure And Realization Of Digital Filters.

UNIT-II DISCRETE TIME FOURIER TRANSFORM(DTFT) & DISCRETE FOURIER TRANSFORM (DFT):

DTFT--Frequency Domain Representation of Discrete Time Signals and Systems. DFT-- Properties of The DFS, DFS Representation Of Periodic Sequences, Properties Of DFT. Convolution Of Sequences, Long Duration Sequence Filtering.

UNIT-III FAST – FOURIER TRANSFORMS (FFT):

Radix – 2 Decimation – In – Time (DIT) And Decimation – In – Frequency (DIF) FFT Algorithms, Inverse FFT.

UNIT-IV IIR DIGITAL FILTER DESIGN TECHNIQUES:

General Considerations in Digital Filter Design. IIR Filter Design-Bilinear Transformation Method, Impulse Invariance Technique. Design Of IIR Filters From Analog Filters (Butterworth Approximation Only). Frequency Transformations.

UNIT-V FIR DIGITAL FILTER DESIGN:

Linear Phase FIR filters, Fourier Series Method, Design Of FIR Filter Using Windows(Rectangular,Bartlett,Hanning& Hamming Windows). Comparison Of IIR And FIR Filters.

Text Books:

1. Alan V. Oppenheim & Ronald W. Schaffer: Digital Signal Processing 1st Edition published by Pearson Education.
2. S. K. Mitra, "Digital Signal Processing: A computer based approach", McGraw Hill, 2011.
3. J. G. Proakis and D.G. Manolakis, "Digital Signal Processing: Principles, Algorithms And Applications", Prentice Hall, 1997.

Reference Books:

1. L. R. Rabiner and B. Gold, "Theory and Application of Digital Signal Processing", Prentice Hall, 1992.
2. J. R. Johnson, "Introduction to Digital Signal Processing", Prentice Hall, 1992.
3. D. J. DeFatta, J. G. Lucas and W. S. Hodgkiss, "Digital Signal Processing", John Wiley & Sons, 1988.
4. Raddae & Rabiner, Application Of Digital Signal Processing.
5. S. P. Eugene Xavier, Signals, Systems And Signal Processing, S. Chand & Co. Ltd.
6. Antonio, Analysis And Design Of Digital Filters, Tata Mc Graw Hill.
7. P. Ramesh Babu: Digital Signal Processing, 4th Edition, Scitech Publications.
8. Moon & Stirling, Mathematical Methods And Algorithms For Signal Processing, Prentice Hall, 2000.

POWER SYSTEM PROTECTION

Lecture : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

1. To know the need for protection, various devices for protection and to be familiar with the terminology used in the power system protection.
2. To understand the working of various types of fuses and circuit breakers and various challenges during their operation.
3. To know the principle of operation, classification, constructional features and testing of circuit breakers.
4. To be familiar with the principle of operation and different types of protective relays and their application to alternators, transformers and feeders.
5. To understand the concepts of over voltage protection and devices used for overvoltage protection.

Course Outcomes:

Students are able to

1. Identify the need for protection and know various devices for protection and terminology used in protection.
2. Discriminate the constructional details with operation principle of various types of fuses, circuit breakers, relays, lightning arresters and their applications.
3. Apply the arc quenching methods to various types of circuit breakers.
4. Apply various relays to various types of power system equipment like alternator, transformer and feeders and distinguish between an electromagnetic relay and a static relay.
5. Identify the different causes for over voltages and choose various protection devices against over voltages.

SYLLABUS

UNIT-I: Introduction to Power System Protection: Need for protective systems, Nature and causes of faults, Types and effects of faults, Fault statistics, Evolution of protective relays, Zones of protection, Primary and Back-up protection, Essential qualities of protection, Classification of protective relays, Classification of Protective Scheme, CTs and PTs and their applications in protection schemes, Summation transformer, Phase-sequence current segregating network, Basic relay terminology.

UNIT-II: Fuses and Circuit Breakers: Fuses and their types, High-voltage HRC fuses and their applications, Selection of fuses. Circuit breakers, Formation of arc, Methods of arc extinction, Restriking voltage, Recovery voltage, RRRV, Single and double frequency transients, Resistance switching, Current chopping, Switching of capacitor banks and unloaded lines, Ratings and characteristics of circuit breakers.

UNIT-III: Types of Circuit Breakers and Testing: Principle of operation of circuit breakers, Classification of circuit breakers, Constructional Features of Air Circuit Breakers, Oil Circuit Breakers, Air Blast Circuit Breakers, SF-6 Circuit Breakers and Vacuum Circuit Breakers, Testing of Circuit Breakers.

UNIT-IV: Protective Relays: Different types of protective relays, Principle of operation and characteristics of relays, Overcurrent, Earth fault and Phase fault protection, Differential and Distance protection with simple applications to Alternators; Transformers; Single and parallel feeders. Introduction to Static relaying, Static relays for time lag Overcurrent and Differential Protection.

UNIT-V: Overvoltage Protection: Causes of overvoltages, Overvoltages due to Lightning, Protection against Lightning and Travelling Waves – Earth Wire, Spark Gap, Surge Arresters, Lightning Arresters, Surge Absorber, Peterson Coil, Insulation Co-ordination.

Text Books:

1. “Power System Protection and Switchgear” by Badri Ram and D. N. Vishwakarma, Tata McGraw-Hill Education, 2001.

Reference Books:

1. “Electrical Power Systems” by C. L. Wadhwa, New Age International, 2009.
2. “Power System Protection” by Paul M Anderson, Wiley, 1998.
3. “Switchgear Protection and Power Systems”, by Sunil S. Rao, Khanna Publishers, 2008.

OOPS THROUGH JAVA
(Common to ECE & EEE)
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Understanding the OOP's concepts, classes and objects, threads, files, applets, swings and act.
2. This course introduces computer programming using the JAVA programming language with object- oriented programming principles.
3. Emphasis is placed on event-driven programming methods, including creating and manipulating objects, classes, and using Java for network level programming and middleware development

Course Outcomes:

1. Understand Java programming concepts and utilize Java Graphical User Interface in Program writing.
2. Write, compile, execute and troubleshoot Java programming for networking concepts.
3. Build Java Application for distributed environment.
4. Design and Develop multi-tier applications.
5. Identify and Analyze Enterprise applications

SYLLABUS**UNIT-I:**

Introduction to OOP, procedural programming language and object oriented language, principles of OOP, applications of OOP, history of java, java features, JVM, program structure.

Variables, primitive data types, identifiers, literals, operators, expressions, precedence rules and associativity, primitive type conversion and casting, flow of control.

UNIT-II:

Classes and objects, class declaration, creating objects, methods, constructors and constructor overloading, garbage collector, importance of static keyword and examples, this keyword, arrays, command line arguments, nested classes.

UNIT-III:

Inheritance, types of inheritance, super keyword, final keyword, overriding and abstract class. Interfaces, creating the packages, using packages, importance of CLASSPATH and java.lang package. Exception handling, importance of try, catch, throw, throws and finally block, user-defined exceptions, Assertions.

UNIT-IV:

Multithreading: introduction, thread life cycle, creation of threads, thread priorities, thread synchronization, communication between threads. Reading data from files and writing data to files, random access file,

UNIT-V:

Applet class, Applet structure, Applet life cycle, sample Applet programs. Event handling: event delegation model, sources of event, Event Listeners, adapter classes, inner classes.

AWT: introduction, components and containers, Button, Label, Checkbox, Radio Buttons, List Boxes, Choice Boxes, Container class, Layouts, Menu and Scrollbar.

Text Books:

1. The complete Reference Java, 8th edition, Herbert Schildt, TMH.
2. Programming in JAVA, Sachin Malhotra, Saurabh Choudary, Oxford.
3. Introduction to java programming, 7th edition by Y Daniel Liang, Pearson.

Reference Books:

1. Swing: Introduction, JFrame, JApplet, JPanel, Componets in Swings, Layout Managers in
2. Swings, JList and JScrollPane, Split Pane, JTabbedPane, JTree, JTable, DialogBox.

UNIX & SHELL PROGRAMMING (Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Written technical communication and effective use of concepts and terminology.
2. Facility with UNIX command syntax and semantics.
3. Ability to read and understand specifications, scripts and programs.
4. Individual capability in problem solving using the tools presented within the class.
5. Students will demonstrate a mastery of the course materials and concepts within in class discussions.

Course Outcomes:

1. Documentation will demonstrate good organization and readability.
2. File processing projects will require data organization, problem solving and research.
3. Scripts and programs will demonstrate simple effective user interfaces.
4. Scripts and programs will demonstrate effective use of structured programming.
5. Scripts and programs will be accompanied by printed output demonstrating completion of a test plan.
6. Testing will demonstrate both black and glass box testing strategies.
7. Project work will involve group participation.

SYLLABUS

UNIT-I

Introduction to unix-Brief History-What is Unix-Unix Components-Using Unix-Commands in Unix-Some Basic Commands-Command Substitution-Giving Multiple Commands.

UNIT-II

The File system –The Basics of Files-What’s in a File-Directories and File Names-Permissions-I Nodes-The Directory Hierarchy, File Attributes and Permissions-The File Command knowing the File Type-The Chmod Command Changing File Permissions-The Chown Command Changing the Owner of a File-The Chgrp Command Changing the Group of a File.

UNIT-III

Using the Shell-Command Line Structure-Met characters-Creating New Commands-Command Arguments and Parameters-Program Output as Arguments-Shell Variables- -More on I/O Redirection-Looping in Shell Programs.

UNIT-IV

Filters-The Grep Family-Other Filters-The Stream Editor Sed-The AWK Pattern Scanningand processing Language-Good Files and Good Filters.

UNIT-V

Shell Programming-Shell Variables-The Export Command-The Profile File a Script Run During Starting-The First Shell Script-The read Command-Positional parameters-The \$? Variable knowing the exit Status-More about the Set Command-The Exit Command-Branching Control Structures-Loop Control Structures-The Continue and Break Statement-The Expr Command: Performing Integer Arithmetic-Real Arithmetic in Shell Programs-The

here Document(<<)-The Sleep Command-Debugging Scripts-The Script Command-The Eval Command-The Exec Command.

Text Books:

1. Introduction to Unix Shell Programming by M.G.Venkateshmurthy, Parson.
2. Unix programming environment by Brain W. Kernighan & Rob Pike, Pearson.

Reference Books:

1. Unix and shell programmingby B.M. Harwani, OXFORD university press.

NEURAL NETWORKS & FUZZY LOGIC (Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students will

1. Understand concept of neural network.
2. Study different types of neural networks and training algorithms.
3. Study fuzzy logic and design of fuzzy logic controllers.
4. Learn applications of ANN & Fuzzy logic techniques in electrical engineering.

Course Outcomes:

Students are able to:

1. Understand neural networks and analyze different types of neural networks.
2. Design training algorithms for neural networks.
3. Analyze and design fuzzy logic systems.
4. Apply AI Techniques in electrical engineering.

SYLLABUS

UNIT – I: INTRODUCTION TO NEURAL NETWORKS

Introduction, Humans and Computers, Biological Neural Networks, Historical development of neural network, Terminology and Topology, Biological and artificial neuron models, Basic learning laws.

UNIT- II:FEED FORWARD NEURAL NETWORKS

Introduction, Perceptron models: Discrete, continuous and multi-category, Training algorithms: Discrete and Continuous Perceptron Networks, Perceptron convergence theorem, Limitations and applications of the Perceptron model, Generalized delta learning rule, Feedforward recall and error back propagation training, Hopfield networks.

UNIT -III: ASSOCIATIVE MEMORIES

Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory), Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem.

UNIT IV:CLASSICAL AND FUZZY SETS

Introduction to classical sets - properties, operations and relations; Fuzzy sets, membership, Uncertainty, operations, properties, fuzzy relations, cardinalities, membership functions. Fuzzy Logic System Components-Fuzzification, Membership value assignment, development of rule base and decision making system, defuzzification to crisp sets, defuzzification methods.

UNIT V ANN AND FUZZY LOGIC APPLICATIONS:

ANN: Load forecasting-System identifications-pattern recognition. Fuzzy logic: Fuzzy logic controller in ALFC system and Fuzzy classification in power system fault transients.

Text Books:

1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications by Rajasekharan and Pai – PHI Publication, 2003.
2. Introduction to Neural Networks using MATLAB 6.0 - S.N.Sivanandam, S.Sumathi, S.N.Deepa, TMH, 2006.

Reference Books:

1. Neural Networks – James A Freeman and Davis Skapura, Pearson Education, 2002.
2. Neural Networks – Simon Hakens , Pearson Education, 3rd edition.
3. Neural Engineering by C.Eliasmith and CH.Anderson, PHI
4. Neural Networks and Fuzzy Logic System by Bart Kosko, PHI Publications, Jan 1992.

VLSI DESIGN
(Common to ECE & EEE (Open Elective))

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3Hrs	Credits	: 3

Course Objectives: Student will be introduced to

1. Use mathematical methods and circuit analysis models in analysis of CMOS digital electronics circuits, including logic components and their interconnections. Learn the various fabrication steps of NMOS and CMOS.
2. Apply CMOS technology-specific layout rules in the placement and routing of transistors and interconnect and to verify the functionality, timing, power and parasitic effects.
3. Learn some basic electrical properties of MOSFET and scaling models and limitations of scaling of MOS circuits.
4. The concepts and techniques of modern integrated circuit design and testing (CMOS VLSI). Learn basic concepts of FPGA.
5. Introduction to Low power CMOS Logic circuits and also some optimisation techniques.

Course Outcomes: By the end of the course the learners (students) will be able to

1. Apply the Concept of design rules during the layout of a circuit.
2. Model and simulate digital VLSI systems using hardware design language.
3. Synthesize digital VLSI systems from register-transfer or higher level descriptions
4. Understand current trends in semiconductor technology, and how it impacts scaling and performance.
5. Understand the basic concepts of FPGA and low power VLSI design

SYLLABUS

UNIT-I: Introduction :

Introduction to IC Technology, Fabrication process: NMOS, PMOS and CMOS. I_{ds} versus V_{ds} Relationships, Aspects of MOS transistor Threshold Voltage, MOS transistor Transconductance, Output Conductance and Figure of Merit. NMOS Inverter, Pull-up to Pull-down Ratio for NMOS inverter driven by another NMOS Inverter, and through one or more pass transistors, Alternative forms of pull-up, The CMOS Inverter, Latch-up in CMOS circuits, Comparison between CMOS and Bi-CMOS technology.

UNIT-II: MOS and Bi-CMOS Circuit Design Processes:

MOS Layers, Stick Diagrams, Design Rules and Layout, General observations on the Design rules, $2\mu\text{m}$ Double Metal, Double Poly, CMOS/BiCMOS rules, $1.2\mu\text{m}$ Double Metal, Double Poly CMOS rules, Layout Diagrams of NAND and NOR gates and CMOS inverter, Symbolic Diagrams-Translation to Mask Form.

UNIT-III: Basic Circuit Concepts:

Sheet Resistance, Sheet Resistance concept applied to MOS transistors and Inverters, Area Capacitance of Layers, Standard unit of capacitance, The Delay Unit, Inverter Delays, Driving large capacitive loads, Propagation Delays, Wiring Capacitances, Choice of layers

Scaling of MOS Circuits: Scaling models, Scaling factors for device parameters, Limits due to sub threshold currents, current density limits on logic levels and supply voltage due to noise and current density. Switch logic, Gate logic.

UNIT-IV: Test and Testability:

Design for Testability, Practical design for Test (OFT) Guidelines, Scan Design Techniques and Built-In-Self Test.

FPGA Based Systems: Introduction, Basic concepts, FPGA architecture.

UNIT-V: Introduction to Low Power VLSI Design:

Introduction to Deep submicron digital IC design, Low power CMOS Logic circuits: Overview of power consumption, Low –Power design through voltage scaling, Estimation and optimisation of switching activity, Reduction of switching capacitance, interconnect Design, Power Grid and Clock Design.

Text Books:

1. Essentials of VLSI Circuits and Systems By Kamran Eshraghian, Douglas and A. Pucknell and Sholeh Eshraghian, Prentice-Hall of India Private Limited, 2005 Edition.
2. CMOS Digital Integrated Circuits Analysis and Design, Sung-Mo Kang, Yusuf Leblebici, Tata McGraw Hill Education, 2003.

Reference Books:

1. “FPGA Based System Design”- Wayne Wolf, Pearson Education, 2004, Technology and Engineering.

INDUSTRIAL ROBOTICS
(Common to ECE & EEE)
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives:

1. To give students practice in applying their knowledge of mathematics, science, and Engineering and to expand this knowledge into the vast area of robotics.
2. The students will be exposed to the concepts of robot kinematics, Dynamics, Trajectory planning.
3. Mathematical approach to explain how the robotic arm motion can be described.
4. The students will understand the functioning of sensors and actuators.

Course Outcomes:

Upon successful completion of this course you should be able to:

1. Identify various robot configuration and components,
2. Select appropriate actuators and sensors for a robot based on specific application
3. Carry out kinematic and dynamic analysis for simple serial kinematic chains.
4. Perform trajectory planning for a manipulator by avoiding obstacles

SYLLABUS

UNIT-I

Introduction: Automation and Robotics, CAD/CAM and Robotics – An over view of Robotics – present and future applications – classification by coordinate system and control system.

UNIT – II

Components Of The Industrial Robotics: Function line diagram representation of robot arms, common types of arms. Components, Architecture, number of degrees of freedom – Requirements and challenges of end effectors, determination of the end effectors, comparison of Electric, Hydraulic and Pneumatic types of locomotion devices.

UNIT – III

Motion Analysis: Homogeneous transformations as applicable to rotation and translation – problems.

Manipulator Kinematics: Specifications of matrices, D-H notation joint coordinates and world coordinates Forward and inverse kinematics – problems.

Differential transformation and manipulators, Jacobians– problems Dynamics: Lagrange – Euler and Newton – Euler formulations – Problems.

UNIT IV

General considerations in path description and generation. Trajectory planning and avoidance of obstacles, path planning, Skew motion, joint integrated motion –straight line motion – Robot programming, languages and software packages-description of paths with a robot programming language..

UNIT V

Robot Actuators and Feed Back Components:

Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors.

Feedback components: position sensors – potentiometers, resolvers, encoders – Velocity sensors.

Robot Applications in Manufacturing:

Material Transfer - Material handling, loading and unloading- Processing - spot and continuous arc welding & spray painting - Assembly and Inspection.

Text Books:

1. Industrial Robotics / Groover M P / Pearson Edu.
2. Robotics and Control / Mittal R K & Nagrath I J / TMH.

Reference Books:

1. Robotics / Fu K S / McGrawHill.
2. Robotic Engineering / Richard D. Klafter, PrenticeHall
3. Robot Analysis and Control / H. Asada and J.J.E. Slotine / BSP Books Pvt.Ltd.
4. Introduction to Robotics / John J Craig / Pearson Edu.

ELECTRICAL MACHINES-II LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int.Marks : 50****Ext. Marks : 50****Credits : 2**

Course Objectives:

Students will conduct experiments on characteristics of synchronous and induction machines.

Course Outcomes:

Students are able to

1. Calculate the regulation of an alternator by EMF, MMF and ZPF methods.
2. Verify Alternator synchronism and draw the performance characteristics, finding out different reactances.
3. Find the efficiency and machine performances by conducting various tests on 3- Φ and 1- Φ induction motor.
4. Verify the speed variation of induction machine.

List of Experiments

1. No Load and Blocked Rotor test on a three-phase squirrel cage induction motor.
2. Regulation of alternator by EMF and MMF methods.
3. Regulation of alternator by ZPF method.
4. Characteristics of line excited induction generator.
5. Characteristics of induction start synchronous motor.
6. Load test on three phase slip ring induction motor.
7. V and inverted V curves of synchronous motor.
8. Measurement of X_d and X_q of a synchronous machine.
9. Equivalent circuit of a single-phase induction motor.
10. Measurement of sequence reactances of a synchronous machine.

Text Books:

1. Nagrath & Kothari, "Electric Machines" TMH 5th edition
2. PS Bhimbra, "Electrical Machinery", Khanna Publishers. 7th edition.

Reference Books:

1. Fitzgerald & Kingsley, "Electric Machinery" McGraw Hill, 6 edition .
2. Alexander S. Langsdorf, "AC Machines", Tata McGraw Hill. 2nd edition

CONTROL SYSTEMS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course objectives:

Students will

1. Understand the basic knowledge on practical control system.
2. Acquire knowledge on applications of machines & electronic devices with control systems.

Course outcomes:

Students are able to

1. Formulate transfer function for given control system problems.
2. Find time response of given control system model.
3. Apply Root Locus and Bode plots for given control system model
4. Design Lead, Lag, Lead-Lag and PID controllers for given control system model

List of experiments:

1. Magnetic amplifier
2. Study of DC Servo motor
3. DC Position control system
4. Study of first order system
5. Study of second order system
6. Speed torque characteristics of AC Servomotor
7. PID Controller
8. Synchro Transmitter and Receiver pair
9. Study of digital control system
10. Study of Lead-Lag compensators

Reference Books:

1. Benjamin C. Kuo, 'Automatic Control Systems', PHI (5th Edition).
2. Katsuhiko Ogata, 'Modern Control Engineering', PHI (4th Edition).
3. Richard C. Dorf and Robert H. Bishop, 'Modern Control Systems', Addison-Wesley Publishers(8th Edition)

EMPLOYABILITY SKILLS
(Common to all Branches)

Theory	: 3 Periods (VA-2+QA-1)	Int.Marks	: 30
Exam	: 3 Hrs.	Ext.Marks	: 70
		Credits	: 1

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II: Puzzle test, blood Relations, permutations, Combinations and probability Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III: Periods,Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

ADVANCED CODING
(Common to ECE & EEE)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 1

Course Objectives

1. To understand the basics of modular programming
2. To learn about ADT, Linked Lists and Templates.
3. To investigate different methods to find time complexities.
4. To learn about Java collections and Libraries

Course Outcomes

At the end of the course, a student should be able to:

1. Acquire coding knowledge on essential of modular programming
2. Acquire Programming knowledge on linked lists
3. Acquire coding knowledge on ADT
4. Acquire knowledge on time complexities of different methods
5. Acquire Programming skill on Java libraries and Collections

SYLLABUS

UNIT I Review Coding essentials and modular programming

Introduction to Linear Data, Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding.

Introduction to modular programming: Formation of methods, Methods: Signature and definition, Inter-method communication, Data casting & storage classes, Recursions

UNIT II Linear Linked Data

Introduction to structure pointer, Creating Links Basic problems on Linked lists, Classical problems on linked lists. Circular Linked lists, Operations on CLL, Multiple links, Operations on Doubly linked lists

UNIT III Abstract Data-structures

Stack data-structure, Operations on stack, Infix/Prefix/Post fix expression evaluations, Implementation of stack using array, Implementation of stack using linked lists.

Queue data-structure: Operations on Queues, Formation of a circular queue, Implementation of queue using stack, Implementation of stack using array, Implementation of stack using linked lists

UNIT IV Running time analysis of code and organization of linear list data

Code evaluation w.r.t running time, Loop Complexities, Recursion complexities, Searching techniques: sequential Vs. binary searching.

Organizing the list data, Significance of sorting algorithms, Basic Sorting Techniques: Bubble sort, selection sort, Classical sorting techniques: Insertion sort, Quick sort, Merge sort.

UNIT V Standard Library templates and Java collections

Introduction to C++ language features, Working on STLs, Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.

Note: This course should focus on Problems

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by RobotLaphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>